BUILDING BLOCKS ESSENTIALS · BUILDING BLOCKS OF ML & APPLIED AI

Building Blocks Handbook

Building Blocks · Class 1 to Class 8 (Excel, RDBMS, DDL, SELECT & joins, SQL Server setup, Statistics 1–3) and Building Blocks ML · Class 1 to Class 10 (programming, Statistics 1–4, Mathematics 1–5), with diagrams, real-world examples and the 2025–26 updates.

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

Interactive version (every tab clickable):
https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw

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COURSE HANDBOOK 01

Building Blocks Essentials

Class 1 to Class 8: Excel, RDBMS, DDL, SELECT & joins, SQL Server setup, Statistics parts 1–3

Building Blocks Essentials · 8 classes · modules: Excel Pre-Learning, SQL Pre-Learning, Building Blocks Assessment

Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#m1**

Overview

Building Blocks Essentials is the foundation course of the programme. It teaches the three skills every data role uses daily: **Excel** for quick analysis, **SQL** for pulling data out of databases, and **statistics** for reading numbers correctly. Every later course (Python, Data Visualization, ML) assumes you know this material.



Where Building Blocks fits: Excel and SQL power the "collect, clean, explore" steps; statistics powers "explore" and "model".

Class	Title	What it covers
Building Blocks · Class 1	Microsoft Excel and basic functionality	Excel environment, terminology, key features, shortcuts; Number, Date & Time, Text and Aggregate functions; formatting and custom number formats; conditional formatting (incl. custom rules); freeze panes; sorting and filters; basic charts
Building Blocks · Class 2	Introduction to RDBMS	Server, database, DBMS, RDBMS; database design and structure; key RDBMS vendors; ER diagrams; keys and constraints; relationships and metadata; introduction to SQL; MS SQL Server and SSMS
Building Blocks · Class 3	Data Definition Language	Create, drop and select databases; create and drop tables; data types; inserting values; UPDATE; ALTER; primary and foreign keys with constraints; TRUNCATE; best practices
Building Blocks · Class 4	Data extraction with SELECT	Subsetting, sorting, filtering, removing duplicates; operators; order of operations; text, date, number and aggregate functions; joins (LEFT, RIGHT, INNER, OUTER, SELF, CROSS, NATURAL); appending; set operators; subqueries; business use cases
Building Blocks · Class 5	SQL recitation: SQL Server setup	Installing SQL Server and SSMS, the query editor, importing flat files
Building Blocks · Class 6	Statistics Part 1/4	What statistics is, data types and scales, population vs sample, central tendency

Class	Title	What it covers
Building Blocks · Class 7	Statistics Part 2/4	Dispersion, percentiles, shape, box plots, z-scores, correlation
Building Blocks · Class 8	Statistics Part 3/4	Probability, distributions, sampling, Central Limit Theorem

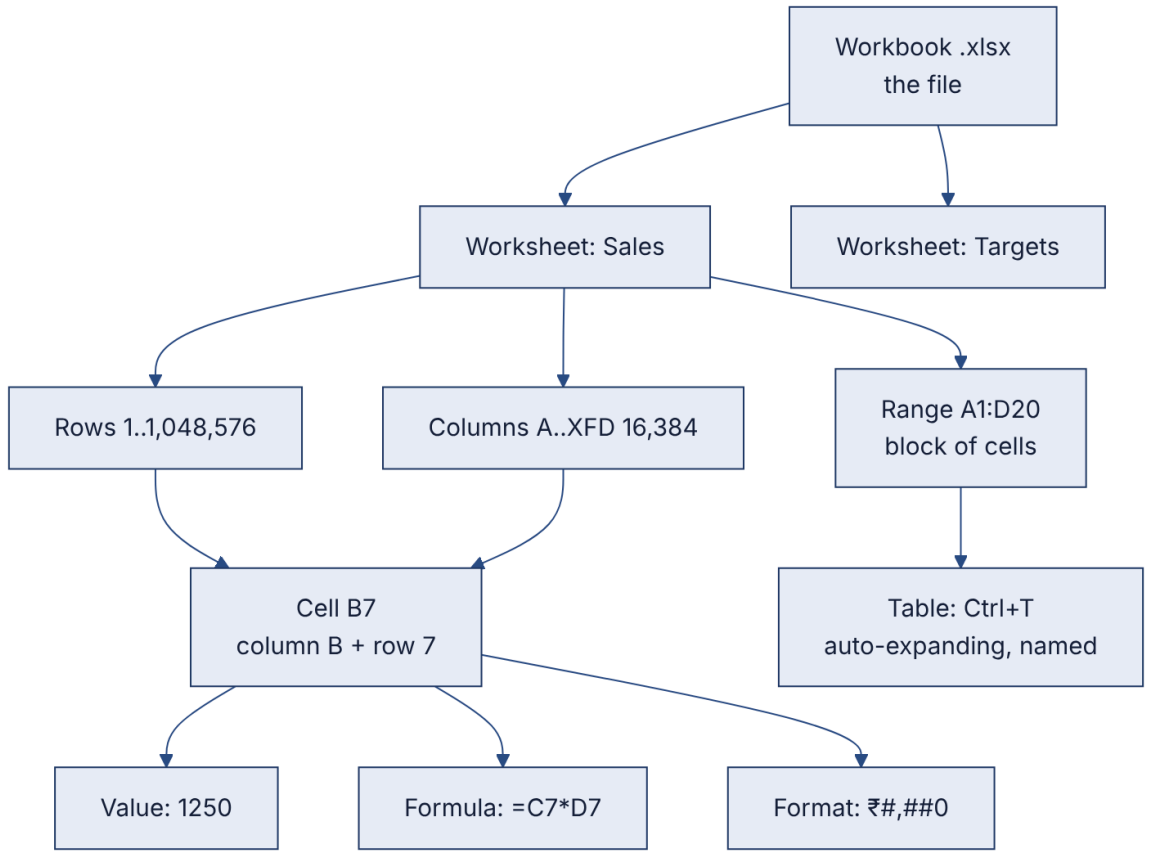
The course page labels Classes 6–8 as "pre-learning sessions on STATISTICS (Parts 1–3 of 4) that should be covered before statistics in Python for Data Science". Part 4/4 is taught in **Building Blocks of ML & Applied AI · Class 5**. The class titles do not list sub-topics, so the split of statistics topics across Parts 1–4 below follows the standard order of an introductory statistics course.

Tip Also see **What's new 2025–26** for the Excel and SQL Server changes released after these classes were recorded, and the **Advanced Excel** and **SQL** learning sessions for the full basic-to-advanced path.

Building Blocks · Class 1

Building Blocks · Class 1: Introduction to Microsoft Excel and basic functionality

1.1 The Excel environment



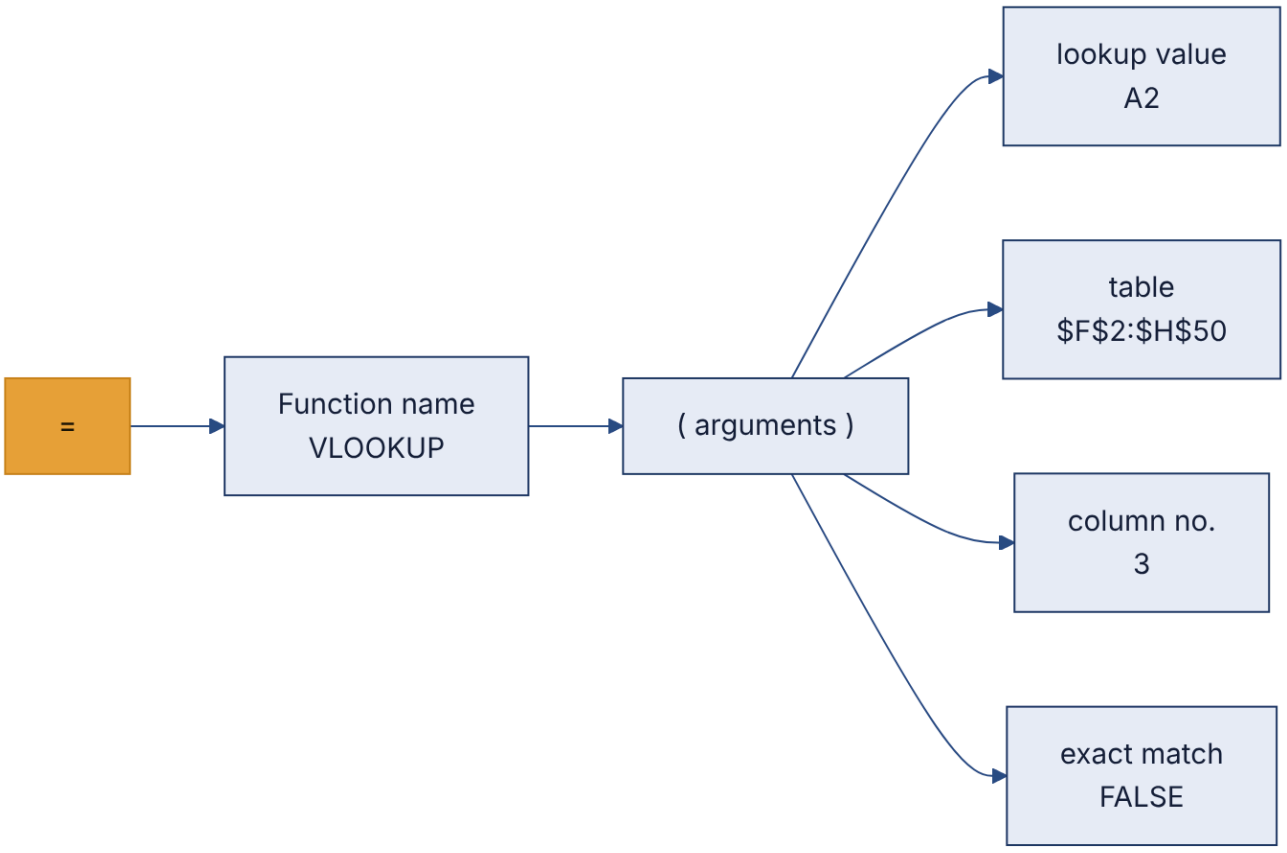
Workbook → worksheets → rows × columns → cells. A modern worksheet has 1,048,576 rows and 16,384 columns (A to XFD).

Part of the screen	What it does
Ribbon	Tabs (Home, Insert, Page Layout, Formulas, Data, Review, View) holding groups of commands
Quick Access Toolbar	Your favourite commands (Save, Undo); right-click any command → Add to QAT
Name Box	Shows the active cell address; type a name or address and press Enter to jump there
Formula Bar	Shows the real content of a cell: the formula, not just the result
Status bar	Select numbers to see instant Sum, Average and Count; right-click it to add Min/Max
Sheet tabs	Switch, rename (double-click), colour, move or copy sheets
File → Options	Calculation mode, autosave, language, add-ins (Analysis ToolPak)

1.2 Terminology

Workbook	The file (.xlsx; .xlsm if it has macros; .xlsb binary; .csv plain text, one sheet, no formatting)
Worksheet	One grid inside the workbook
Cell / address	Intersection of a column and row, e.g. C7
Range	A block of cells, e.g. A1:D20; non-adjacent ranges use a comma: A1:A5,C1:C5

Formula	Starts with `=` ; can use operators, references and functions: `=B2*C2`
Function	A built-in formula with a name and arguments: `=SUM(B2:B10)`
Argument	An input to a function, separated by commas
Relative / absolute reference	A1 shifts when copied; $A$1 stays fixed; A$1 / $A1 are mixed. Press `F4` to cycle
Table	A range converted with `Ctrl` + `T` : banded, auto-expanding, uses names like `Sales[Amount]`



Anatomy of a formula: equals sign, function name, then arguments in brackets.

1.3 Key functionalities

- **AutoFill and Flash Fill**: drag the fill handle to continue a series (Jan, Feb...); `Ctrl` + `E` Flash Fill learns a pattern from one example (first names from full names).
- **Paste Special** (`Ctrl` + `Alt` + `V`): values only, formats, transpose, multiply a range by a number.
- **Find & Replace** (`Ctrl` + `H`), Go To Special (blanks, formulas, errors).
- **Text to Columns** (Data tab): split "Gurugram, Haryana" into two columns.
- **Remove Duplicates**, **Data Validation** (drop-down lists), **Protect Sheet**.

1.4 Shortcuts

Navigate and select	
`Ctrl` + `Arrow`	Jump to the last filled cell in a direction
`Ctrl` + `Shift` + `Arrow`	Select to the edge of data
`Ctrl` + `Home` / `End`	Go to A1 / last used cell
`Ctrl` + `Space` / `Shift` + `Space`	Select column / row
`Ctrl` + `PgUp/PgDn`	Previous / next sheet
`Ctrl` + `G` or `F5`	Go To

Edit and format	
`F2`	Edit cell
`F4`	Toggle $ references / repeat last action
`Alt` + `=`	AutoSum
`Ctrl` + `1`	Format Cells
`Ctrl` + `Shift` + `$ % #`	Currency / percent / date format
`Ctrl` + `;` / `Ctrl` + `Shift` + `;`	Today's date / current time
`Alt` + `Enter`	New line inside a cell
`Ctrl` + `Shift` + `L`	Toggle filters
`Ctrl` + `` ` ``	Show formulas instead of results

1.5 Functions: four families with real examples

Example sheet: an electronics retailer's orders. Columns: A Order ID, B Order Date, C Customer, D City, E Product, F Qty, G Unit Price, H Amount.

Family	Function	Real-world formula	Result / meaning
Number	ROUND	`=ROUND(H2*18%,2)`	GST at 18%, rounded to paise
	ROUNDUP / ROUNDDOWN	`=ROUNDUP(F2/12,0)`	Cartons needed when 12 fit per carton
	INT / TRUNC	`=INT(H2/1000)`	Whole thousands
	MOD	`=MOD(A2,2)=0`	TRUE for even order IDs (alternate-row checks)
	ABS	`=ABS(Actual-Target)`	Size of a miss, ignoring sign
	CEILING / FLOOR, RAND, RANDBETWEEN	`=CEILING(G2,10)`	Price rounded up to the next ₹10

Family	Function	Real-world formula	Result / meaning
Date & Time	TODAY / NOW	`=TODAY()-B2`	Days since the order
	YEAR / MONTH / DAY	`=YEAR(B2)`	Order year for grouping
	TEXT	`=TEXT(B2,"mmm-yy")`	"Oct-26" month label
	DATE	`=DATE(2026,4,1)`	Start of the Indian financial year
	EOMONTH / EDATE	`=EOMONTH(B2,0)`	Month-end date (billing cut-off); `EDATE(B2,3)` = 3 months later
	DATEDIF	`=DATEDIF(DOB,TODAY(),"Y")`	Age in completed years (hidden function, still works)
	WEEKDAY / NETWORKDAYS / WORKDAY	`=NETWORKDAYS(B2,DeliveredOn)`	Working days to deliver (excludes weekends, optional holiday list)
Text	LEFT / RIGHT / MID	`=LEFT(A2,3)`	Region prefix "NOR" from "NOR-1042"
	LEN	`=LEN(Phone)=10`	Valid 10-digit mobile number?
	FIND / SEARCH	`=MID(Email,FIND("@",Email)+1,50)`	Email domain
	TRIM / CLEAN	`=TRIM(C2)`	Removes extra spaces from imported names
	UPPER / LOWER / PROPER	`=PROPER(C2)`	"ravi kumar" → "Ravi Kumar"
	CONCAT / & / TEXTJOIN	`=TEXTJOIN(", ",TRUE,D2:D6)`	One cell listing cities, skipping blanks
	SUBSTITUTE / REPLACE	`=SUBSTITUTE(G2,"₹","")*1`	Strip a symbol and convert to a number
	VALUE / TEXT	`=VALUE("1,250")`	Text that looks like a number → a real number

Family	Function	Real-world formula	Result / meaning
Aggregate	SUM / AVERAGE	`=SUM(H2:H500)`	Total revenue
	COUNT / COUNTA / COUNTBLANK	`=COUNTBLANK(D2:D500)`	Orders missing a city
	MIN / MAX / SMALL / LARGE	`=LARGE(H2:H500,3)`	Third-largest order
	SUMIF / SUMIFS	`=SUMIFS(H:H,D:D,"Pune",B:B,">="&DATE(2026,4,1))`	Pune revenue this financial year
	COUNTIF / COUNTIFS	`=COUNTIFS(D:D,"Delhi",H:H,">10000")`	Big Delhi orders
	AVERAGEIF(S), MEDIAN, MODE	`=AVERAGEIFS(H:H,E:E,"Laptop")`	Average laptop order value

1.6 Formatting and custom number formatting

Formatting changes how a value **looks**, never the value itself. 0.18 formatted as % shows 18% but is still 0.18 in formulas.

Custom format (Ctrl+1 → Custom)	Value	Shows as	Use
`#,##0`	1250000	1,250,000	Thousands separators
`[>=10000000]##\,##\,##\,##0;[>=100000]##\,##\,##0;##,##0`	1250000	12,50,000	Indian lakh/crore grouping
`0.0,,"M"`	1250000	1.3M	Millions in dashboards (each comma divides by 1,000)
`"₹"#,##0.00`	99.5	₹99.50	Currency
`000`	7	007	Fixed-length IDs
`dd mmm' yyyy`	09-05-1986	09 May' 1986	Readable dates
`0 "kg"`	80	80 kg	Units while staying numeric
`[Green]▲0%;[Red]▼0%;0%`	0.12 / −0.05	▲12% / ▼5%	Positive;negative;zero sections
`;;;`	anything	(hidden)	Hide values but keep them

> **Tip** A format has up to four sections separated by semicolons: **positive;negative;zero;text**. Conditions in square brackets, e.g. `[<100000]`, can replace the default sections.

1.7 Conditional formatting and custom rules

- **Highlight Cells Rules**: greater than, between, duplicate values, text contains.
- **Top/Bottom**: top 10 items, top 10%, above average.

- **Data Bars, Colour Scales, Icon Sets**: in-cell visuals for a quick heatmap.
- **Custom rule with a formula** (New Rule → Use a formula): the formula must return TRUE/FALSE for the **first** cell of the selection; lock columns with $.

```
Highlight the whole row when Amount > 50,000      select A2:H500 →  =$H2>50000
Overdue invoices (due date passed, unpaid)        =AND($E2<TODAY(),$F2="Unpaid")
Weekend orders                                     =WEEKDAY($B2,2)>5
Every other row (zebra)                            =MOD(ROW(),2)=0
Duplicate customer IDs                             =COUNTIF($A$2:$A$500,$A2)>1
```

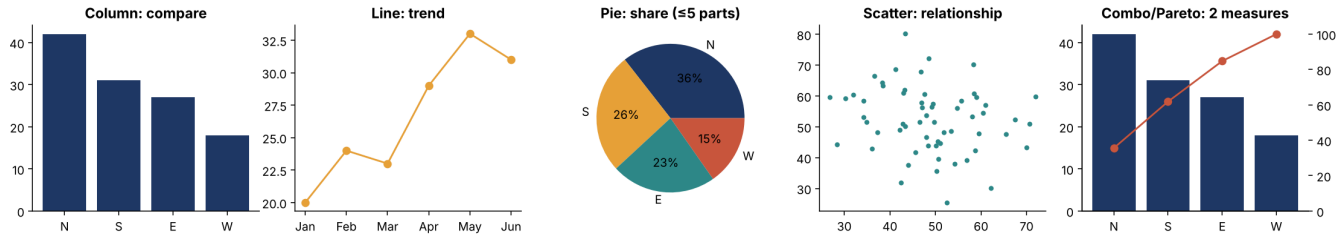
1.8 Freezing panes

View → Freeze Panes keeps headers visible while scrolling. **Freeze Top Row** or **First Column**, or select cell B2 and choose Freeze Panes to lock both row 1 and column A (everything above and left of the selected cell is frozen). Split (View → Split) shows two parts of a sheet side by side.

1.9 Sorting and filters

- Sort A→Z / Z→A on one column; **Custom Sort** for multiple levels (City A→Z, then Amount largest first) or by cell colour, or by a custom list (Jan, Feb, Mar...).
- Filter (`Ctrl` + `Shift` + `L`): pick values, Text/Number/Date filters (Top 10, Between, This Month), filter by colour.
- **Advanced Filter** (Data → Advanced): criteria range with AND (same row) / OR (different rows); copy unique records to another location.
- Use `SUBTOTAL(9,H2:H500)` or `AGGREGATE` to sum only visible (filtered) rows.

1.10 Charts: concept and creation



Pick the chart from the question: compare, trend, share, relationship, or two measures together.

1. Select the data, including headers (or click inside a Table).
2. Insert → Recommended Charts, or choose Column/Line/Pie (`Alt` + `F1` inserts a default chart instantly; `F11` on a new sheet).
3. Use the **+** button for chart elements: title, axis titles, data labels, legend.
4. Chart Design → Change Chart Type → Combo for two measures (revenue bars + margin % line on a secondary axis).

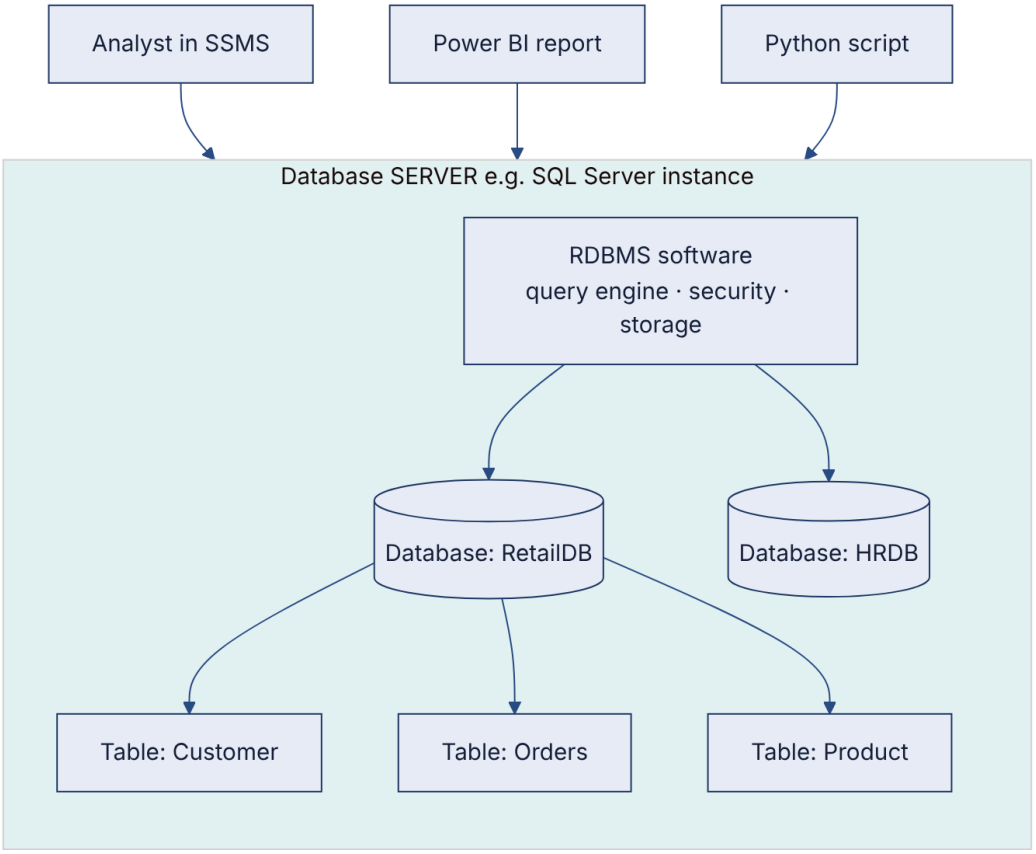
5 Format: remove clutter (gridlines, 3D), sort bars, write the insight in the title ("South leads with 34% of sales").

Tip Practise every Class 1 topic in the **Excel Basic-to-Advanced workbook**: one sheet per topic with live formulas.

Building Blocks · Class 2

Building Blocks · Class 2: Introduction to RDBMS

2.1 Server, database, DBMS, RDBMS



Users and tools connect to a server; the RDBMS software on it manages databases, and each database holds related tables.

Server	A computer (physical or cloud) that runs database software and answers requests over a network. In SQL Server one machine can host several *instances*.
Database (DB)	An organised collection of related data, e.g. RetailDB with Customer, Orders and Product tables.
DBMS	Software to create, read, update and secure data (handles storage, concurrency, backup, permissions).

RDBMS	A DBMS that stores data in **tables (relations)** of rows and columns linked by keys, and is queried with SQL. Follows ACID properties: Atomicity, Consistency, Isolation, Durability.

Excel sheet vs database table

- Excel: ~1M rows per sheet, one user editing, any value in any cell.
- Database: billions of rows, thousands of users, every column has a fixed data type and rules.

Real example

When you order on a food app, the order row is written to an Orders table, linked by customer_id to Customer and by restaurant_id to Restaurant. Analysts later query those same tables for reports.

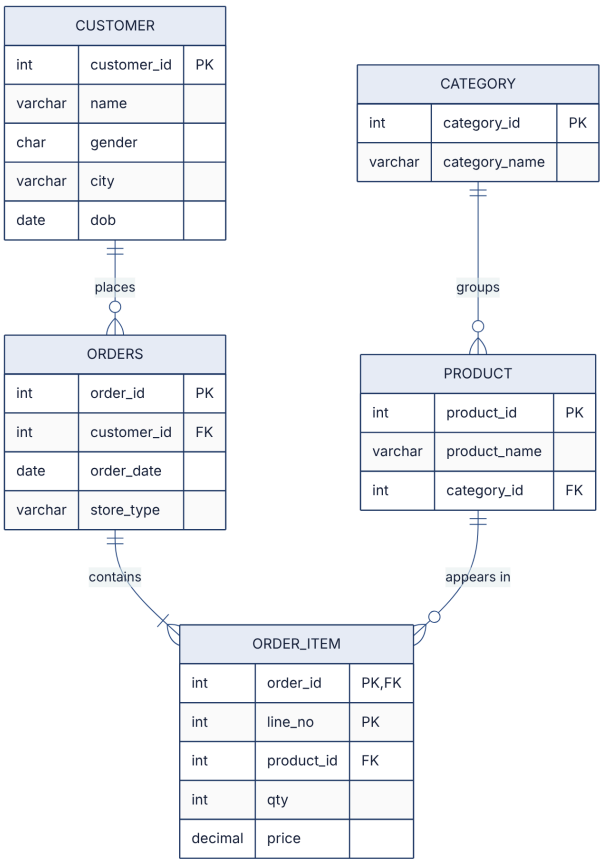
2.2 Database design and structure

- **Table** (relation/entity) → **row** (record/tuple) → **column** (field/attribute).
- **Schema**: a named container of objects (dbo.Customer, sales.Orders).
- **Normalisation**: split data so each fact is stored once. 1NF: one value per cell, no repeating groups. 2NF: every non-key column depends on the whole key. 3NF: no column depends on another non-key column (store city once in a City table, not in every order).
- **OLTP vs OLAP**: transactional systems (normalised, fast writes) vs analytics warehouses (star schemas, fast reads).

2.3 Key RDBMS vendors

Product	Vendor	Notes
SQL Server / Azure SQL	Microsoft	Used in this course; T-SQL dialect; free Developer and Express editions
MySQL	Oracle (open source)	Popular for web apps
PostgreSQL	Open source	Feature-rich; very popular in start-ups and cloud
Oracle Database	Oracle	Large enterprises, banking
SQLite	Open source	File-based, inside phones and apps
Cloud warehouses	Snowflake, Google BigQuery, Amazon Redshift, Databricks SQL, Microsoft Fabric	Analytics at scale; standard SQL with small dialect differences

2.4 Entity Relationship Diagram (ERD)



Crow's-foot ERD for a retail database. "||--o{" reads "exactly one ... zero or many".

2.5 Keys and constraints

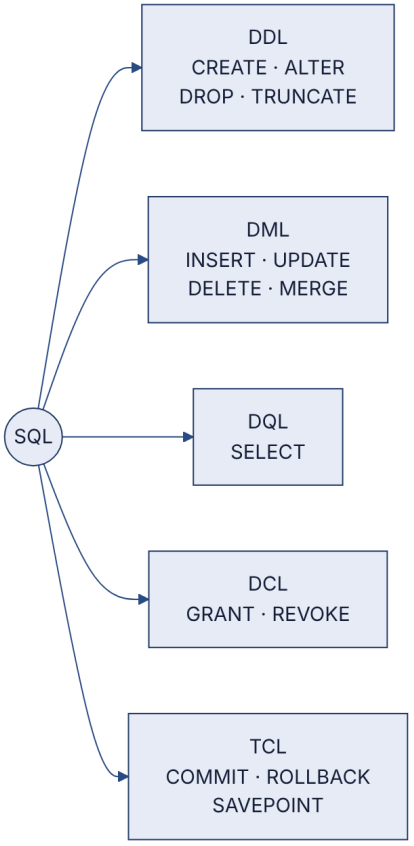
Key / constraint	Meaning	Example
Primary key (PK)	Uniquely identifies each row; not NULL; one per table	customer_id
Foreign key (FK)	References a PK in another table; enforces referential integrity	Orders.customer_id → Customer.customer_id
Candidate key	Any minimal column set that could be the PK	customer_id, email, PAN
Alternate key	A candidate key not chosen as PK	email
Composite key	PK made of 2+ columns	(order_id, line_no)
Super key	Any unique column set, even with extra columns	(customer_id, name)
Surrogate vs natural key	System-generated ID vs a real-world identifier	IDENTITY(1,1) vs PAN number
NOT NULL · UNIQUE · CHECK · DEFAULT	Column rules	`CHECK (qty > 0)`, `DEFAULT GETDATE()`

2.6 Relationships and metadata

- **One-to-one**: employee ↔ employee_passport. **One-to-many**: customer → orders (most common). **Many-to-many**: students ↔ courses, implemented with a bridge table Enrolment(student_id, course_id).
- **Metadata** is data about data: table names, column names, types, keys, row counts. Query it with `SELECT * FROM INFORMATION_SCHEMA.COLUMNS WHERE TABLE_NAME = 'Customer';` or `EXEC sp_help 'Customer';`

2.7 SQL, SQL Server and SSMS

SQL (Structured Query Language) is the standard language for relational databases. It is **declarative**: you say *what* you want; the engine decides *how*. **T-SQL** is Microsoft's dialect (adds variables, TOP, procedures). **SSMS** (SQL Server Management Studio) is the free Windows tool for connecting to SQL Server, browsing objects in Object Explorer and running queries.



The five SQL sub-languages.

Building Blocks · Class 3

Building Blocks · Class 3: Data Definition Language

3.1 Create, select and drop a database

```
CREATE DATABASE RetailDB;           -- create
USE RetailDB;                       -- select it for the following statements
SELECT name FROM sys.databases;     -- list databases
-- DROP DATABASE RetailDB;          -- deletes the database and ALL its data (irreversible)
```

3.2 Data types

Category	Types	Use for
Whole numbers	TINYINT (0–255), SMALLINT, INT (±2.1 billion), BIGINT	IDs, counts
Exact decimals	DECIMAL(p,s) / NUMERIC, MONEY	Money: DECIMAL(10,2) = up to 99,999,999.99
Approximate	FLOAT, REAL	Scientific values (never for money)
Text	CHAR(n) fixed, VARCHAR(n) variable, NVARCHAR(n) Unicode (Hindi, emojis), VARCHAR(MAX)	Names, codes, comments
Date & time	DATE, TIME, DATETIME2, DATETIMEOFFSET (with time zone)	Order dates, timestamps
Other	BIT (0/1), UNIQUEIDENTIFIER, VARBINARY, JSON (SQL Server 2025), VECTOR (2025)	Flags, GUIDs, files, documents, AI embeddings

3.3 Create and drop tables, with keys and constraints

```
CREATE TABLE Customer (
    customer_id INT IDENTITY(1001,1) PRIMARY KEY,          -- auto number 1001, 1002...
    full_name   NVARCHAR(80) NOT NULL,
    email       VARCHAR(120) UNIQUE,
    city        VARCHAR(40)  DEFAULT 'Gurugram',
    dob         DATE CHECK (dob < '2010-01-01')
);
CREATE TABLE Orders (
    order_id    INT PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date  DATE NOT NULL DEFAULT CAST(GETDATE() AS DATE),
    amount      DECIMAL(10,2) CHECK (amount >= 0),
    CONSTRAINT fk_orders_customer FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)
);
-- DROP TABLE Orders;   (drop the child first: Customer cannot be dropped while Orders references
it)
```

3.4 Insert values

```
INSERT INTO Customer (full_name, email, city, dob)
VALUES ('Asha Verma', 'asha@example.com', 'Pune', '1994-03-12'),
       ('Ravi Kumar', 'ravi@example.com', DEFAULT, '1989-11-02');     -- multi-row insert (max 1000
rows per VALUES)
INSERT INTO Orders VALUES (1, 1001, '2026-10-01', 2499.00);           -- all columns in table
order
INSERT INTO Orders_Archive SELECT * FROM Orders WHERE order_date < '2026-01-01';   -- insert from a
query
```

3.5 Modify records: UPDATE (always with WHERE)

```sql
UPDATE Customer SET city = 'Noida' WHERE customer_id = 1002;
UPDATE Orders   SET amount = amount * 1.05 WHERE order_date >= '2026-10-01';   -- 5% price revision
-- Without WHERE every row changes. Run the same WHERE as a SELECT first to check the rows.
```

3.6 ALTER: change the table's properties

```sql
ALTER TABLE Customer ADD phone VARCHAR(15);                      -- add a column
ALTER TABLE Customer ALTER COLUMN city VARCHAR(60);              -- change type/size
ALTER TABLE Customer DROP COLUMN phone;                          -- remove a column
ALTER TABLE Orders ADD CONSTRAINT chk_amount CHECK (amount < 1000000);
ALTER TABLE Orders DROP CONSTRAINT chk_amount;
EXEC sp_rename 'Customer.full_name', 'customer_name', 'COLUMN';  -- rename a column
```

3.7 DELETE vs TRUNCATE vs DROP

	DELETE	TRUNCATE TABLE	DROP TABLE
Removes	Rows matching WHERE (or all)	All rows, keeps structure	Rows + structure + constraints
Language	DML	DDL	DDL
Identity counter	Continues	Resets to seed	Gone
Works with FK references	Yes (if rules allow)	Not if referenced by a FK	Not while referenced
Rollback in a transaction	Yes	Yes in SQL Server (inside BEGIN TRAN)	Yes in SQL Server (inside BEGIN TRAN)

3.8 Best practices and key takeaways

✔ Name consistently: singular or plural tables, snake_case columns, `pk_` / `fk_` / `chk_` constraint prefixes
✔ Choose the smallest correct type: DATE not VARCHAR for dates, DECIMAL for money
✔ Every table gets a primary key; every relationship gets a foreign key
✔ List columns in INSERT; never rely on column order
✔ Wrap risky changes in a transaction: `BEGIN TRAN … ROLLBACK / COMMIT`
✔ Keep scripts in files (and Git) so the database can be rebuilt

Building Blocks · Class 4

Building Blocks · Class 4: Data extraction with SELECT

Examples use the retail tables: Customer(customer_id, name, gender, city, dob), Orders(order_id, customer_id, order_date, store_type, amount), Product(product_id, product_name, category, price), Order_Item(order_id, product_id, qty).

4.1 Subsetting, sorting, filtering, removing duplicates

```sql
SELECT * FROM Customer;                                    -- all columns (avoid in production)
SELECT name, city FROM Customer;                           -- subset columns
SELECT TOP 10 order_id, amount FROM Orders ORDER BY amount DESC;   -- top 10 orders
SELECT DISTINCT city FROM Customer;                        -- remove duplicates
SELECT * FROM Orders WHERE store_type = 'e-Shop' AND amount > 5000;
SELECT * FROM Orders ORDER BY order_date DESC, amount ASC;         -- multi-level sort
```

4.2 Operators

Type	Operators	Example
Assignment	`=` in SET and variables	`DECLARE @fy DATE = '2026-04-01';`
Arithmetic	`+ - * / %`	`amount * 0.18 AS gst`; integer division: `7/2 = 3`, use `7/2.0`
Comparison (conditional)	`= <> != > < >= <=`	`WHERE amount <> 0`
Range / list	`BETWEEN … AND`, `IN (…)`	`WHERE city IN ('Pune','Delhi')`
Wildcards with LIKE	`%` any chars, `_` one char, `[a-c]` set, `[^a]` not	`WHERE name LIKE 'A%'`; `LIKE '_a%'`
NULL handling	`IS NULL`, `IS NOT NULL`, `ISNULL()`, `COALESCE()`	`COALESCE(phone, email, 'no contact')`
Logical	`AND OR NOT`	`WHERE NOT (city = 'Delhi' OR amount < 100)`

Tip Precedence: brackets → arithmetic → comparison → NOT → AND → OR. `WHERE a OR b AND c` means `a OR (b AND c)`; add brackets to be clear.

4.3 Sequence of operations



SQL is written SELECT-first but executed FROM-first. That is why a SELECT alias cannot be used in WHERE, but can be used in ORDER BY.

4.4 Functions

Group	Functions	Business example
Text	LEN, LEFT, RIGHT, SUBSTRING, CHARINDEX, UPPER, LOWER, TRIM, REPLACE, CONCAT, STRING_AGG	`UPPER(LEFT(city,3))` city code; `STRING_AGG(product_name, ', ')` products per order
Date & time	GETDATE, YEAR, MONTH, DATEPART, DATENAME, DATEADD, DATEDIFF, EOMONTH, FORMAT, CONVERT	`DATEDIFF(DAY, order_date, GETDATE())` days since order
Number	ROUND, CEILING, FLOOR, ABS, POWER, SQRT	`ROUND(AVG(amount), 2)`

Group	Functions	Business example
Aggregate	COUNT, COUNT(DISTINCT), SUM, AVG, MIN, MAX	`COUNT(DISTINCT customer_id)` active customers
Conversion / logic	CAST, CONVERT, TRY_CONVERT, CASE, IIF	`CASE WHEN amount >= 10000 THEN 'High' ELSE 'Normal' END`

```sql
SELECT city,
       COUNT(*)                    AS orders,
       SUM(o.amount)               AS revenue,
       ROUND(AVG(o.amount), 0)     AS avg_order_value
FROM Orders o JOIN Customer c ON o.customer_id = c.customer_id
WHERE o.order_date >= '2026-04-01'
GROUP BY city
HAVING SUM(o.amount) > 100000
ORDER BY revenue DESC;
```

4.5 Joins: extracting data from multiple tables

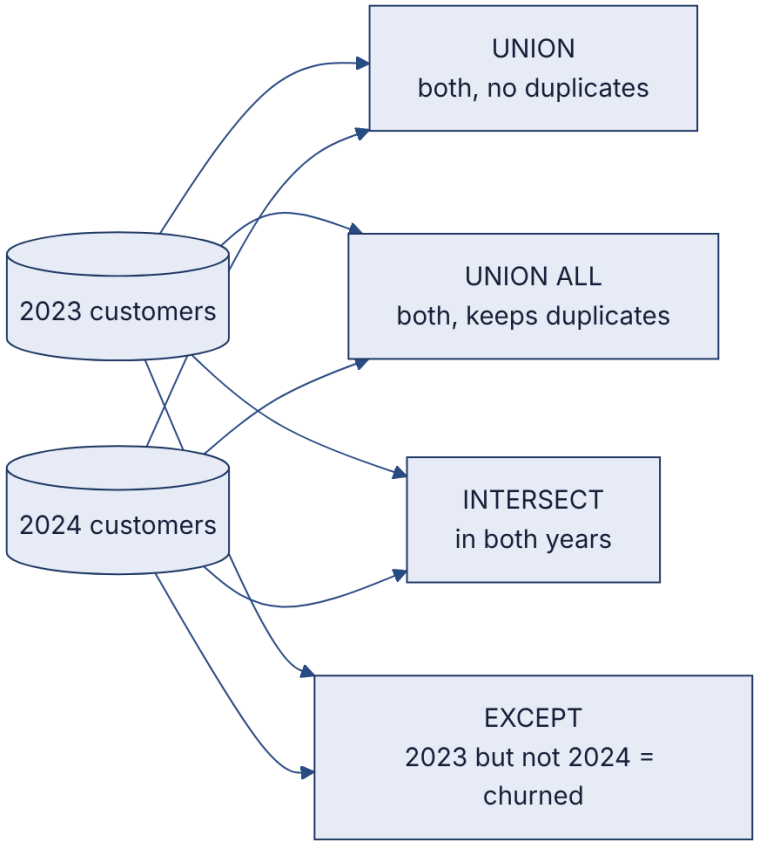


The six join types.

```sql
-- INNER: orders with their customer
SELECT o.order_id, c.name, o.amount FROM Orders o INNER JOIN Customer c ON o.customer_id =
c.customer_id;
-- LEFT: every customer, even those who never ordered (find inactive customers)
SELECT c.name FROM Customer c LEFT JOIN Orders o ON c.customer_id = o.customer_id WHERE o.order_id
IS NULL;
```

```sql
-- RIGHT: mirror of LEFT;  FULL OUTER: unmatched rows from both sides (data reconciliation)
SELECT a.id, b.id FROM System_A a FULL OUTER JOIN System_B b ON a.id = b.id WHERE a.id IS NULL OR
b.id IS NULL;
-- SELF: employee with manager name
SELECT e.name AS employee, m.name AS manager FROM Employee e LEFT JOIN Employee m ON e.manager_id =
m.emp_id;
-- CROSS: every store × every month (to build a complete calendar grid)
SELECT s.store_id, m.month_start FROM Store s CROSS JOIN Months m;
-- NATURAL JOIN joins automatically on same-named columns. It exists in MySQL/PostgreSQL/Oracle
-- but NOT in SQL Server; always write explicit ON conditions instead.
```

4.6 Appending datasets and set operators



Set operators combine rows (vertically); joins combine columns (horizontally). Both queries must have the same number of columns with compatible types.

```sql
SELECT customer_id FROM Orders_2023
UNION ALL SELECT customer_id FROM Orders_2024;      -- append (fast, keeps duplicates)
SELECT customer_id FROM Orders_2023
EXCEPT    SELECT customer_id FROM Orders_2024;      -- churned customers
```

4.7 Subqueries

```sql
-- Ordinary (nested) subquery in WHERE
SELECT * FROM Orders WHERE amount > (SELECT AVG(amount) FROM Orders);
SELECT name FROM Customer WHERE customer_id IN (SELECT customer_id FROM Orders WHERE store_type =
'Flagship store');
```

```sql
-- Inline view (derived table) in FROM
SELECT city, AVG(cust_rev) AS avg_revenue_per_customer
FROM (SELECT c.city, c.customer_id, SUM(o.amount) AS cust_rev
      FROM Customer c JOIN Orders o ON c.customer_id = o.customer_id
      GROUP BY c.city, c.customer_id) AS t
GROUP BY city;
```

4.8 Business use cases

Business question	SQL pattern
Top 5 cities by revenue this FY	JOIN + WHERE date + GROUP BY + ORDER BY + TOP 5
Customers who never ordered	LEFT JOIN ... WHERE right.key IS NULL (or NOT EXISTS)
Repeat customers (more than 3 orders)	GROUP BY customer HAVING COUNT(*) > 3
Share of revenue by store type	SUM(amount) * 100.0 / (SELECT SUM(amount) FROM Orders)
Orders above average	Subquery in WHERE
Return rate by category	SUM(CASE WHEN qty < 0 THEN 1 ELSE 0 END) * 1.0 / COUNT(*)

> **Tip** Full worked practice with real results: **SQL Retail case study** (15 questions) and **SQL Mobile case study**.

Building Blocks · Class 5

Building Blocks · Class 5: SQL recitation – installing SQL Server, SSMS and importing data



From installation to your first query on imported data.

5.1 Install SQL Server

1. Download **SQL Server Developer** (free, full features, non-production) or **Express** (free, small production) from Microsoft's SQL Server downloads page. The current release is SQL Server 2025.
2. Choose **Basic** installation (or Custom to pick the instance name). Note the instance name shown at the end, e.g. `localhost\SQLEXPRESS` or `localhost` for a default instance.
3. Install **SSMS** separately (it is a different download). Recent SSMS versions (21 and later) install through the Visual Studio Installer and add a dark theme and optional Copilot.

5.2 Connect and use the IDE

- Connect dialog: Server type *Database Engine*; Server name from step 2; Authentication *Windows Authentication*. If a certificate warning appears on a local machine, tick *Trust server certificate*.

- **Object Explorer** (left): Databases → Tables → Columns, Keys, Constraints, Indexes. Right-click a table → *Select Top 1000 Rows*.
- **Query editor**: New Query (`Ctrl` + `N`); pick the database in the toolbar drop-down or write `USE RetailDB;`; run with `F5`; run only the highlighted lines by selecting them first; `Ctrl` + `R` hides the results pane; IntelliSense refresh `Ctrl` + `Shift` + `R`.
- Results to Grid (`Ctrl` + `D`) or Text (`Ctrl` + `T`); right-click the grid → Save Results As CSV.

5.3 Import flat files (CSV / TXT)

1. Right-click the database → **Tasks → Import Flat File**.
2. Browse to `Transactions.csv`; table name defaults to the file name.
3. **Preview** the first rows. In **Modify Columns**, set data types (DATE for dates, DECIMAL for amounts), mark a primary key, and allow NULLs where needed.
4. Finish, then check: `SELECT COUNT(*) FROM Transactions;` should match the file's row count.

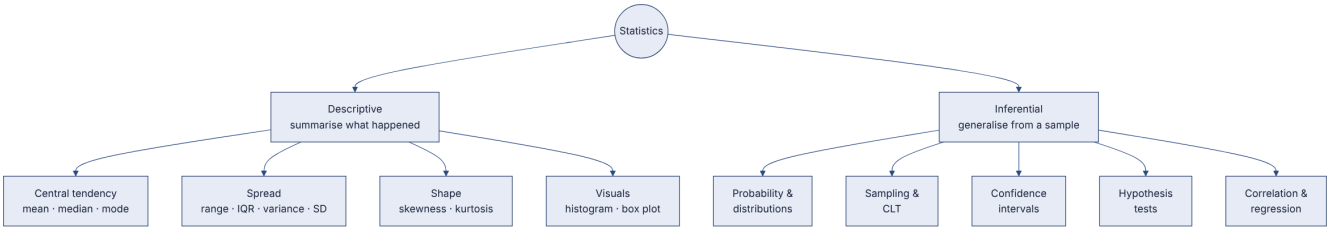
> **Common import errors:** dates in dd-mm-yyyy fail as DATE → import as VARCHAR, then convert with `TRY_CONVERT(DATE, tran_date, 105)`; amounts with commas or ₹ → import as text and clean; text longer than the default nvarchar(50) → widen the column in Modify Columns.

For Excel files use **Tasks → Import Data** (SQL Server Import and Export Wizard), or save the sheet as CSV first. For repeatable loads write `BULK INSERT`:

```
BULK INSERT dbo.Transactions
FROM 'C:\Data\Transactions.csv'
WITH (FORMAT = 'CSV', FIRSTROW = 2, FIELDTERMINATOR = ',', ROWTERMINATOR = '0x0a');
```

Building Blocks · Class 6

Building Blocks · Class 6: Introduction to Statistics (Part 1/4)



Statistics in one picture: descriptive (Parts 1–2) and inferential (Parts 3–4).

6.1 What statistics is and why analysts need it

Statistics is the science of **collecting, summarising and drawing conclusions from data** while measuring uncertainty. Example: an e-commerce firm sees average order value rise from ₹1,240 to ₹1,310 after a new checkout. Statistics answers whether that rise is real or just normal week-to-week noise.

6.2 Population vs sample, parameter vs statistic

	Population	Sample
Meaning	Every unit you care about (all 2 crore customers)	The subset you actually observe (5,000 surveyed customers)
Summary is called	Parameter	Statistic
Mean / SD symbols	μ, σ	$\bar{x}$, s
Size	N	n

6.3 Types of data and measurement scales

Scale	Type	Can you…	Example
Nominal	Categorical	Count, mode only	City, payment mode, gender
Ordinal	Categorical (ordered)	Rank, median	Rating Poor < Average < Good; education level
Interval	Numeric, no true zero	Add/subtract	Temperature in °C, calendar year
Ratio	Numeric, true zero	All maths incl. ratios	Sales, weight, age, delivery time

Numeric data is **discrete** (countable: items in cart) or **continuous** (measurable: weight). Data is also **cross-sectional** (many customers at one time), **time series** (one metric over time) or **panel** (many customers over time); and **structured** (tables) or **unstructured** (reviews, images).

6.4 Measures of central tendency

Measure	How	Excel	Best when
Mean	$\Sigma x / n$	`=AVERAGE()`	Symmetric numeric data
Median	Middle value of sorted data (average of the two middles if n is even)	`=MEDIAN()`	Skewed data: salaries, house prices, delivery times
Mode	Most frequent value	`=MODE.SNGL()`	Categories: most common shoe size
Weighted mean	$\Sigma(w \cdot x) / \Sigma w$	`=SUMPRODUCT(x,w)/SUM(w)`	Average price across products sold in different quantities
Trimmed mean	Mean after dropping top/bottom x%	`=TRIMMEAN(range,0.1)`	Judges' scores, outlier-prone data
Geometric mean	$(x_1 \cdot x_2 \cdot \ldots \cdot x_n)^{(1/n)}$	`=GEOMEAN()`	Average growth rates (CAGR)

Worked example: why the median matters

Salaries of 6 employees (₹ lakh): 4, 5, 5, 6, 7, 45. Mean = 72/6 = **12**; median = (5+6)/2 = **5.5**; mode = **5**. One senior salary pulls the mean far above what a typical employee earns, so report the median.

Building Blocks · Class 7

Building Blocks · Class 7: Introduction to Statistics (Part 2/4)

7.1 Measures of dispersion (spread)

Measure	Formula	Excel	Note
Range	max − min	`=MAX()-MIN()`	Very sensitive to outliers
Variance (sample)	$s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$	`=VAR.S()`	Squared units
Standard deviation	$s = \sqrt{s^2}$	`=STDEV.S()` (population: `STDEV.P`)	Same units as data
Coefficient of variation	$s / \bar{x} \times 100\%$	`=STDEV.S()/AVERAGE()`	Compare consistency of things on different scales
IQR	Q3 − Q1	`=QUARTILE.INC(r,3)-QUARTILE.INC(r,1)`	Robust to outliers

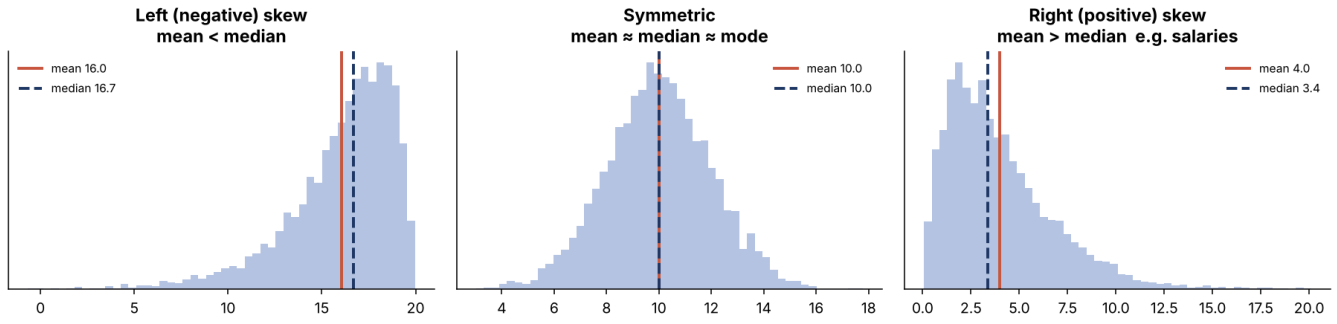
Worked example: two delivery partners

Partner A delivery times (min): 28, 30, 31, 29, 32 → mean 30, SD ≈ 1.6. Partner B: 15, 45, 20, 40, 30 → mean 30, SD ≈ 12.7. Same average, but A is far more reliable. Customers feel the spread, not the mean.

7.2 Percentiles and quartiles

The p-th percentile is the value below which p% of observations fall. Q1 = 25th, Q2 = median, Q3 = 75th percentile. "Your score is at the 90th percentile" means you beat 90% of test-takers. Excel: `=PERCENTILE.INC(range,0.9)`, `=PERCENTRANK.INC(range,x)`.

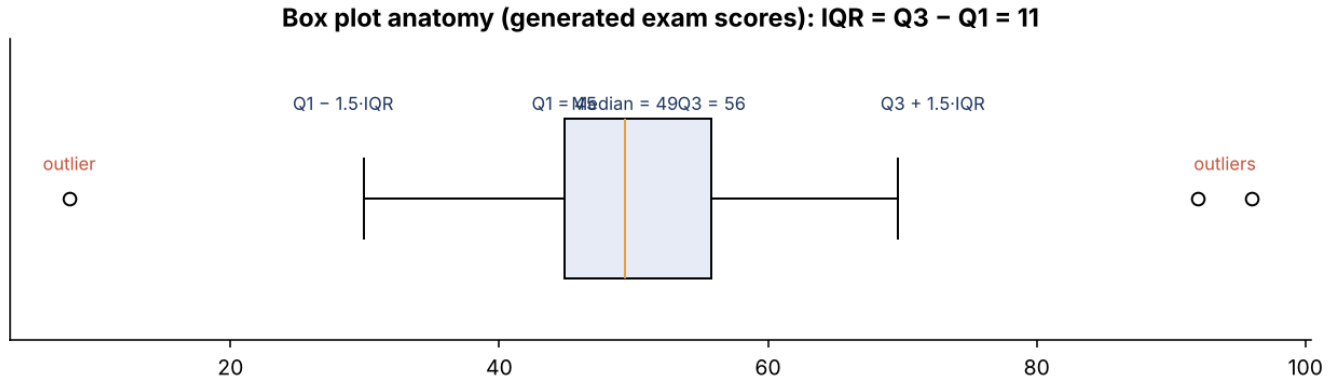
7.3 Shape: skewness and kurtosis



Generated data. In right-skewed data the long tail drags the mean above the median.

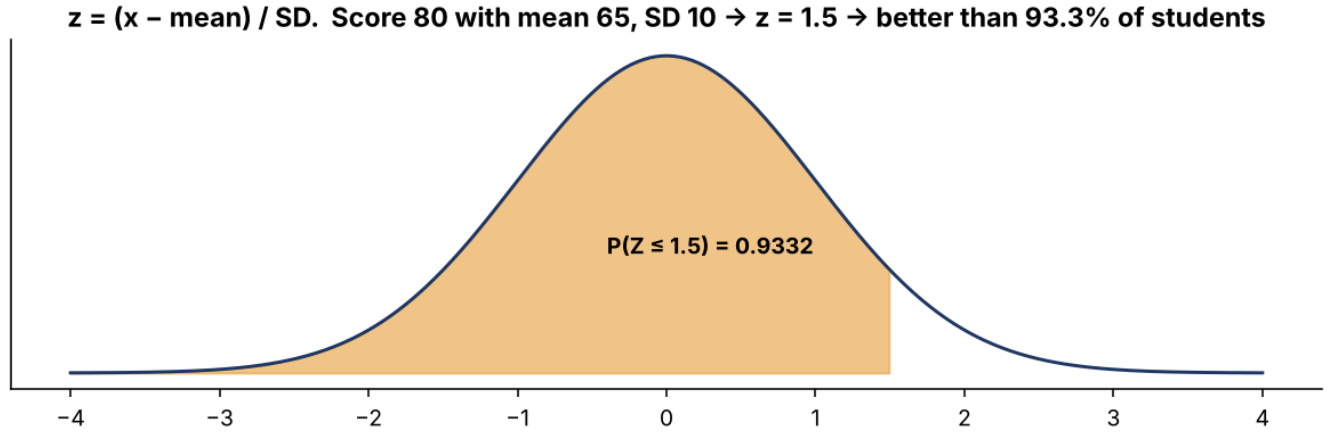
Excel: `=SKEW()` (positive = right tail) and `=KURT()` (positive = heavier tails than normal, more extreme values).

7.4 Box plot and outliers



The box spans Q1–Q3; whiskers reach the last points within 1.5 × IQR; points beyond are flagged as outliers.

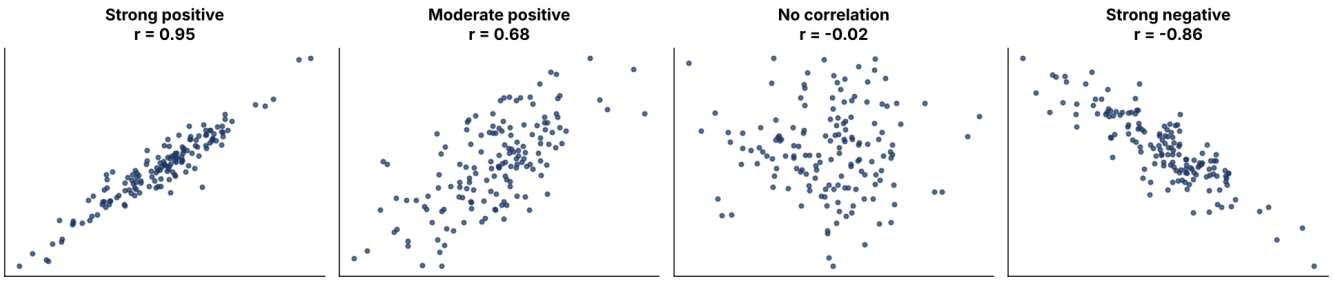
7.5 Standardisation: z-scores



A z-score says how many standard deviations a value is from the mean, so different scales can be compared.

Excel: `=STANDARDIZE(x, mean, sd)`; probability below: `=NORM.S.DIST(z,TRUE)`. Rule of thumb: $|z| > 3$ is unusual.

7.6 Relationship between two variables: covariance and correlation



Pearson's r runs from −1 to +1 and only measures straight-line association.

- Covariance: direction of joint movement, but its size depends on units. `=COVARIANCE.S()`
- Correlation r = cov / ($s_x \cdot s_y$): unit-free. `=CORREL()`. $|r|$ > 0.7 strong, 0.3–0.7 moderate, < 0.3 weak.
- **Correlation is not causation**: ice-cream sales and AC sales both rise in summer because of heat, not because of each other.

7.7 Visualising distributions

Histogram (Insert → Statistic Chart → Histogram; set bin width), box & whisker chart, frequency table with `=FREQUENCY()` or a pivot with grouped values.

Building Blocks · Class 8

Building Blocks · Class 8: Introduction to Statistics (Part 3/4)

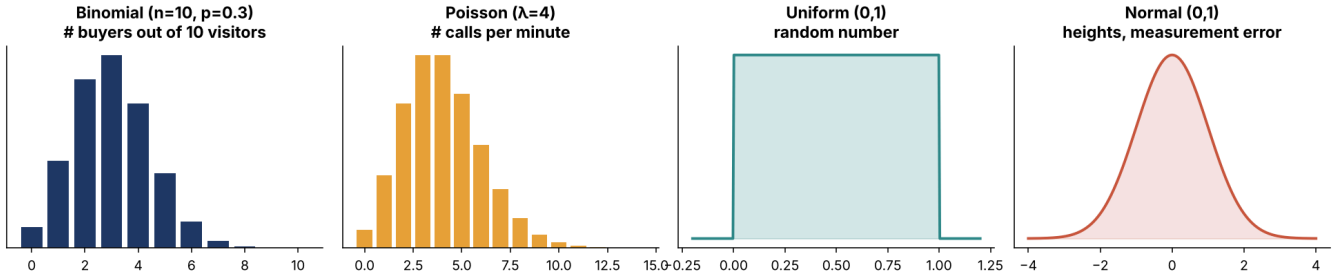
8.1 Probability basics

Experiment, outcome, event	Show an ad (experiment); click/no-click (outcomes); "user clicks" (event)
Probability	P(A) = favourable / total, between 0 and 1. Empirical: 230 clicks out of 10,000 views → P = 0.023
Complement	P(not A) = 1 − P(A)
Addition rule	P(A or B) = P(A) + P(B) − P(A and B)
Multiplication (independent)	P(A and B) = P(A) × P(B): two independent servers both failing
Conditional	P(A | B) = P(A and B) / P(B): probability of purchase given the user added to cart
Bayes' theorem	P(A | B) = P(B | A) · P(A) / P(B)

Worked Bayes example: fraud alert

1% of transactions are fraud. The rule flags 95% of frauds and wrongly flags 5% of genuine ones. If a transaction is flagged, P(fraud | flag) = 0.95×0.01 / (0.95×0.01 + 0.05×0.99) = 0.0095 / 0.059 = **16%**. Most alerts are false alarms because fraud is rare. This is why analysts look at precision, not just detection rate.

8.2 Random variables and probability distributions

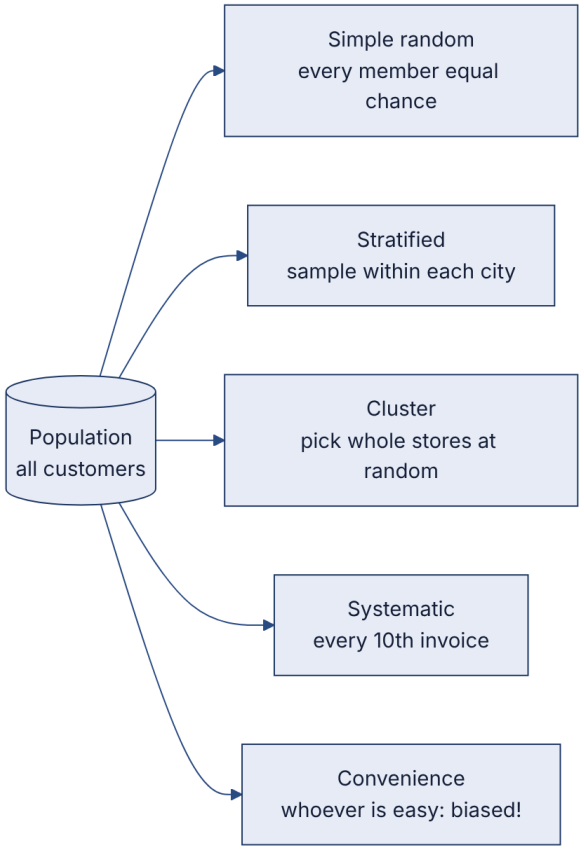


Discrete (bars) vs continuous (curves) distributions with business examples.

Distribution	Models	Excel
Bernoulli / Binomial	Number of successes in n yes/no trials	`=BINOM.DIST(k,n,p,FALSE)`
Poisson	Count of events in a fixed interval (calls per hour)	`=POISSON.DIST(k,λ,FALSE)`
Uniform	All values equally likely	`=RAND()`
Normal	Symmetric bell curve (heights, errors, sums of many effects)	`=NORM.DIST(x,μ,σ,TRUE)`
Exponential	Time between events	`=EXPON.DIST()`
t, χ^2, F	Sampling distributions used by tests (Part 4)	`T.DIST, CHISQ.DIST, F.DIST`

Expected value E[X] = Σ x·P(x): if a lottery ticket wins ₹1,000 with probability 0.001, its expected value is ₹1, so a ₹10 ticket loses ₹9 on average.

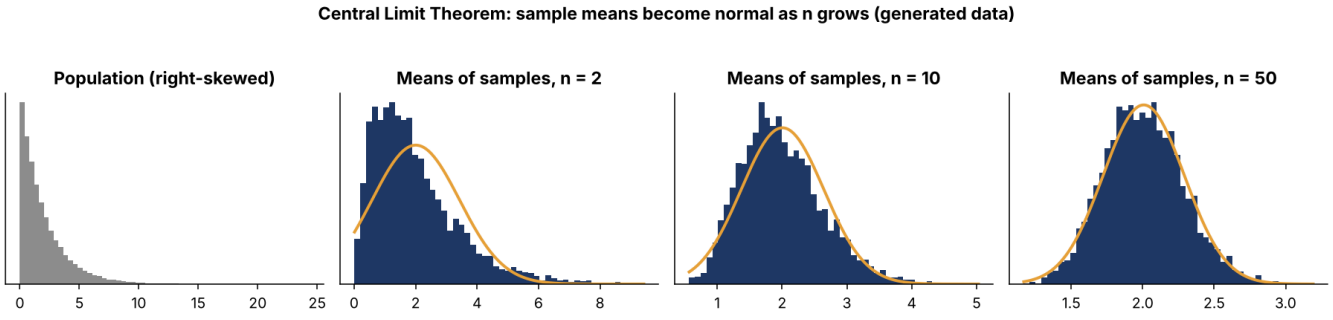
8.3 Sampling methods



Probability methods (random, stratified, cluster, systematic) allow valid inference; convenience samples are biased.

Biases to watch: selection bias (only app users surveyed), non-response bias, survivorship bias (studying only successful start-ups).

8.4 Sampling distribution and the Central Limit Theorem



Means of repeated samples form a normal curve, whatever the population shape, with spread σ/√n (the standard error).

- Bigger samples → smaller standard error → more precise estimates (4× the sample halves the error).
- The CLT is what lets us use z- and t-tests on skewed business data (Part 4).

What's new 2025–26

What changed since these classes were recorded

Excel (Microsoft 365) – new functions and features

Feature	What it does	Example
XLOOKUP / XMATCH	Lookup in any direction with a built-in "not found" value	`=XLOOKUP(A2,Products[ID],Products[Price],"Missing")`
Dynamic arrays: FILTER, SORT, SORTBY, UNIQUE, SEQUENCE	One formula spills a whole result range	`=SORT(FILTER(Sales,Sales[City]="Pune"),4,-1)`
LET / LAMBDA	Name parts of a formula; create your own reusable functions	`=LET(t,SUM(H:H),g,t*18%,t+g)`
TEXTSPLIT, TEXTBEFORE, TEXTAFTER, VSTACK, HSTACK, TAKE, DROP	Modern text and array shaping	`=TEXTBEFORE(A2,"@")`
GROUPBY, PIVOTBY, PERCENTOF	Pivot-table results from a formula	`=GROUPBY(Sales[City],Sales[Amount],SUM)`
REGEXTEST, REGEXEXTRACT, REGEXREPLACE	Pattern matching in text	`=REGEXTEST(B2,"^[6-9]\d{9}$")` valid Indian mobile
TRIMRANGE and trim references (A:.A)	Ignore empty trailing rows in whole-column references	`=SUM(H:.H)`
TRANSLATE, DETECTLANGUAGE	Translate text inside a cell	`=TRANSLATE(A2,"en","hi")`
IMPORTCSV / IMPORTTEXT (2026)	Pull a CSV file into the sheet as a dynamic array	`=IMPORTCSV("C:\data\sales.csv")`
Python in Excel (`=PY()`) + Python editor	Run pandas and charts inside cells (cloud-executed)	`=PY(xl("Sales[#All]",headers=True).describe())`
Copilot in Excel	Generate and explain formulas, fix errors, clean data, analyse text, build PivotTables and charts from a prompt	"Add a column with profit margin and highlight negatives"
`=COPILOT()` function	Calls AI inside a formula, row by row	`=COPILOT("Classify this review as Positive/Negative/Neutral",C2)`

Feature	What it does	Example
Agent Mode	Multi-step tasks (build a model or dashboard) from chat	"Create a monthly sales dashboard from this table"
Other	PivotTable auto-refresh, Show Changes history (one year), dark mode for the grid, Power Query in Excel for the web	

Availability depends on your Microsoft 365 plan, channel and region; Copilot features need a Copilot licence. Excel 2019/2021 standalone versions do not receive these.

SQL Server – newer T-SQL features

Release	Feature	Example
2022	GREATEST / LEAST	`SELECT GREATEST(q1, q2, q3, q4) AS best_quarter`
2022	DATETRUNC, DATE_BUCKET	`DATETRUNC(MONTH, order_date)` first day of the month
2022	GENERATE_SERIES	`SELECT value FROM GENERATE_SERIES(1, 12)` month numbers
2022	IS [NOT] DISTINCT FROM	NULL-safe comparison
2022	WINDOW clause, STRING_SPLIT with ordinal, APPROX_PERCENTILE_CONT	Reuse a window definition
2025	Regular expressions: REGEXP_LIKE, REGEXP_REPLACE, REGEXP_SUBSTR, REGEXP_INSTR, REGEXP_COUNT	`WHERE REGEXP_LIKE(email, '^[^@]+@[^@]+\.[a-z]{2,}$')`
2025	Native JSON data type, JSON indexes, JSON aggregates (JSON_OBJECTAGG, JSON_ARRAYAGG)	Store API responses efficiently
2025	VECTOR data type, VECTOR_DISTANCE, DiskANN vector index (preview)	Similarity search on AI embeddings
2025	Fuzzy string matching (preview), external AI model definitions, change event streaming (preview)	Match "Gurgaon" with "Gurugram"

SQL Server 2025 became generally available in November 2025. Regex and vector functions also exist in Azure SQL Database.

Practice & case study

Real-world mini case: "Mumbai Electronics" monthly review

A 6-store electronics chain shares an Orders export (Order ID, Date, Store, City, Category, Qty, Unit Price, Customer Type). The manager asks five questions. The answer to each uses only Building Blocks skills.

Question	Excel (Class 1)	SQL (Classes 2–4)	Statistics (Classes 6–8)
Revenue by store this month?	`=SUMIFS(Amount,Store,A2,Date,">="&EOMONTH(TODAY(),-1)+1)`	`GROUP BY store WHERE order_date >= DATETRUNC(MONTH, GETDATE())`	Mean and median order value per store
Which store is most consistent?	`=STDEV.S` of daily sales ÷ mean (CV)	`STDEV(daily_total)` in a derived table	Coefficient of variation
Highlight loss-making lines	Conditional formatting `=$J2<0`	`WHERE margin < 0`	Share of loss lines
Are big orders unusual?	`=STANDARDIZE(H2,AVERAGE(H:H),STDEV.S(H:H))`	`(amount - AVG(amount) OVER()) / STDEV(amount) OVER()`	$\|z\| > 3$ → outlier
Probability a walk-in buys a warranty	`=COUNTIFS(Type,"Walk-in",Warranty,"Yes")/COUNTIF(Type,"Walk-in")`	`AVG(CASE WHEN warranty='Yes' THEN 1.0 ELSE 0 END)`	Conditional probability

The **Excel workbook** in Downloads includes a "Case Study – Sales Dashboard" sheet with this scenario built out using live formulas.

More practice on the real course data: **Excel Sportsmen case**, **Excel Consumer Complaints case**, **SQL Retail case**.

Interview Q&A

Q What is the difference between a function and a formula in Excel?

A formula is anything starting with = (=A1+B1). A function is a named, built-in calculation used inside a formula (SUM, VLOOKUP).

Q Relative vs absolute reference?

A1 changes when copied; $A$1 stays fixed; A$1 and $A1 lock only the row or column. Use $ for constants such as a tax-rate cell.

Q How do you sum only visible rows after filtering?

`=SUBTOTAL(109,range)` or `=AGGREGATE(9,5,range)`. SUM includes hidden rows.

Q What is a primary key vs a foreign key?

A primary key uniquely identifies rows in its own table; a foreign key in another table points to that primary key and enforces that the referenced row exists.

Q DELETE vs TRUNCATE?

DELETE removes selected rows and fires triggers; TRUNCATE removes all rows quickly, resets identity, and cannot be used on a table referenced by a foreign key.

Q Why can't I use a column alias in WHERE?

WHERE is evaluated before SELECT, so the alias does not exist yet. Repeat the expression or wrap the query in a subquery/CTE.

Q Mean vs median: which would you report for delivery time?

Median (and the 90th percentile), because delivery times are right-skewed: a few very late orders inflate the mean.

Q What does a standard deviation tell a manager?

How much individual values typically differ from the average. Two stores with equal average sales but different SD differ in predictability.

Q Explain the Central Limit Theorem simply.

If you take many reasonably large samples and compute each sample's average, those averages form a bell curve around the true average, even if the original data is skewed. This makes confidence intervals and tests possible.

Resources

- ▶ YouTube: Simplilearn Excel full course · ▶ YouTube: Simplilearn SQL full course · ▶ YouTube: Simplilearn statistics for data science full course
- ▶ YouTube: Leila Gharani Excel functions you need to know · ▶ YouTube: Chandoo conditional formatting formulas
- ▶ YouTube: Alex The Analyst SQL basics tutorial · ▶ YouTube: install SQL Server and SSMS step by step
- ▶ YouTube: StatQuest statistics fundamentals · ▶ YouTube: Khan Academy statistics and probability
- 📘 Microsoft: What's new in Excel (Tech Community blog) · 📘 Excel functions by category
- 📘 Microsoft Learn: What's new in SQL Server 2025 · 📘 Regex support GA announcement
- 📘 Download SQL Server · 📘 Download SSMS
- 📘 Excel Campus: 17 Excel updates from 2025

COURSE HANDBOOK 02

🧮 Building Blocks of ML & Applied AI

Class 1 to Class 10: programming foundations, Statistics parts 1–4, Mathematics parts 1–5

Building Blocks of ML and Applied AI · 10 classes · modules: Programming Elements, Statistics Pre-Learning, Mathematics Foundation

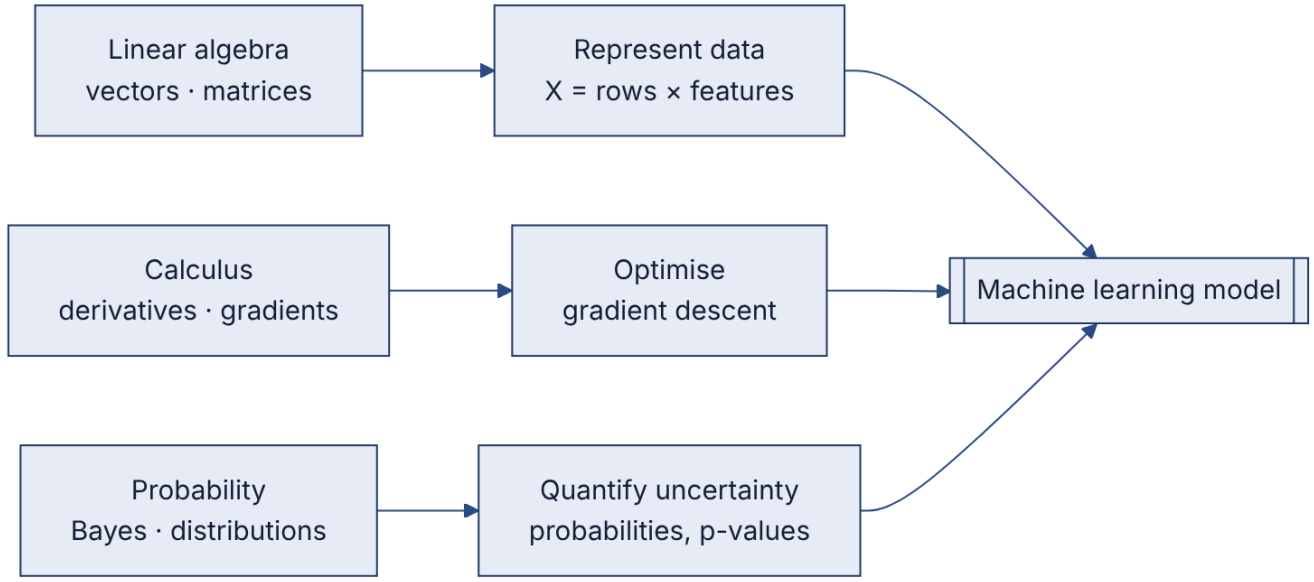
Open this module online: **https://claude.ai/artifact/GtUBqgmq7C6gvyYa7LjhEw#m1b**

Overview

This pre-learning course prepares you for Python, statistics in Python, and the Machine Learning course. Class 1 teaches how programming works; Classes 2–5 are the four statistics pre-learning sessions; Classes 6–10 are the five mathematics sessions needed for machine learning.

Class	Topic	Prepares you for
Building Blocks ML · Class 1	Introduction to programming: programs and business use, databases, IDE, languages, high vs low level, compiler vs interpreter, syntax, variables, naming conventions, data types, operators, control flow, functions and UDFs, logic building and pseudo-code	Python for Data Science
Building Blocks ML · Class 2	Statistics Part 1/4: introduction, data types, central tendency	Python Classes 13–14, 19–21
Building Blocks ML · Class 3	Statistics Part 2/4: dispersion, shape, z-scores, correlation	EDA in Python
Building Blocks ML · Class 4	Statistics Part 3/4: probability, distributions, sampling, CLT	Python Classes 19–20
Building Blocks ML · Class 5	Statistics Part 4/4: estimation, confidence intervals, hypothesis testing, tests, regression basics	Python Class 21, ML Classes 1–3
Building Blocks ML · Class 6	Mathematics Part 1/5: algebra, functions, exponents and logarithms	Link functions, log transforms
Building Blocks ML · Class 7	Mathematics Part 2/5: vectors and matrices	How data is stored for ML
Building Blocks ML · Class 8	Mathematics Part 3/5: matrix operations, inverse, systems of equations, eigenvalues	Linear regression, PCA

Class	Topic	Prepares you for
Building Blocks ML · Class 9	Mathematics Part 4/5: calculus: derivatives, partial derivatives, gradients, optimisation	Gradient descent (ML Class 5)
Building Blocks ML · Class 10	Mathematics Part 5/5: probability for ML: sets, counting, conditional probability, Bayes, expectation	Naive Bayes, model evaluation



The three maths pillars of machine learning.

> The course lists Classes 2–5 as "Statistics Part 1/4 to 4/4" and Classes 6–10 as "Mathematics Part 1/5 to 5/5" without sub-topics. The topic split shown here follows the standard order of these pre-learning syllabi. Statistics Parts 1–3 are the same lectures as **Building Blocks Essentials Classes 6–8**, so those tabs summarise and add practice.

Building Blocks ML · Class 1

Building Blocks ML · Class 1: Introduction to programming

1.1 Computer programs and their business use

A **program** is a precise set of instructions a computer follows to turn **input** into **output**. Businesses use programs to do repeatable work at scale without errors:

Business area	Program does	Example
Banking	Validates and records transactions; flags fraud	UPI payment app checks balance, writes to ledger, sends SMS

Business area	Program does	Example
Retail / e-commerce	Pricing, inventory, recommendations	"Customers who bought this also bought..."
Operations	Scheduling, routing, forecasting	Delivery app assigns the nearest rider
Analytics	Cleans data, builds reports and models every day	Python script that emails a sales report at 9 AM

1.2 Databases and why software needs them

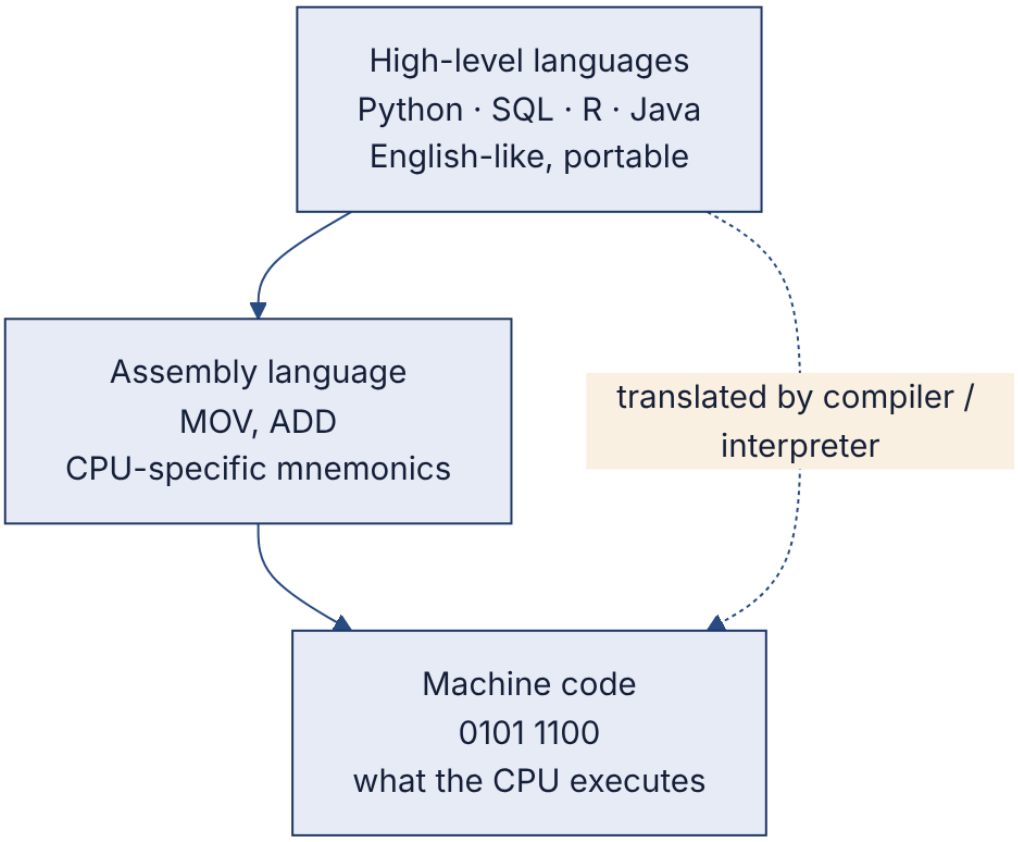
Programs forget everything when they stop. A **database** stores data permanently, lets many users read and write at once safely, enforces rules (an order must have a valid customer), and answers questions quickly with indexes. Almost every application is **front end** (screens) + **back end** (business logic program) + **database**.

1.3 IDE: Integrated Development Environment

An IDE is one application that combines a code **editor** (syntax colours, auto-complete), a way to **run** code, a **debugger** (pause and inspect), and file/project management.

Jupyter Notebook / JupyterLab	Code + output + notes in cells; the standard for data analysis (used in this course)
VS Code	Free general editor with Python and Jupyter extensions, Git, AI assistants
PyCharm, Spyder	Python-focused IDEs (Spyder looks like RStudio/MATLAB)
Google Colab	Free Jupyter in the browser, nothing to install
SSMS	The IDE for SQL Server

1.4 Programming languages; high-level vs low-level



High-level languages read like English and run anywhere; they are translated down to the machine code a CPU executes.

Language	Typical use
Python	Data science, ML, automation, web back ends
SQL	Querying databases
R	Statistics and research
Java / C#	Enterprise applications
JavaScript	Web pages and apps
C / C++	Systems, games, high performance (also inside NumPy)

1.5 Language translators: compiler vs interpreter



A compiler translates the whole program before running; an interpreter translates and runs line by line.

	Compiler	Interpreter
Translates	Entire program at once into an executable	One statement at a time while running
Errors	All reported before running	Stops at the first error it reaches
Speed	Faster execution	Slower, but instant feedback, great for exploring data
Examples	C, C++, Go, Rust	Python, R, JavaScript (Python first compiles to bytecode, then interprets it)

1.6 Why syntax rules?

Computers cannot guess intent. **Syntax** is the grammar of a language (brackets, colons, indentation, quotes). A syntax error means the translator cannot read the code at all; a **logic error** means it runs but gives the wrong answer (much harder to find).

```
if amount > 5000            # SyntaxError: missing colon
    print("High value")

total = price + quantity    # runs, but logic error: should be price * quantity
```

1.7 Programming basics

Variables

A variable is a named label pointing to a value in memory: `city = "Pune"`, `amount = 2499.0`. The value can change; the name lets you reuse it.

INC rules: Identifier Naming Conventions

Rule	Valid	Invalid
Start with a letter or underscore	`total_sales`, `_temp`	`2nd_quarter`
Only letters, digits, underscore	`sales_2026`	`sales-2026`, `sales 2026`, `sales$`
Case-sensitive	`Sales` ≠ `sales`	
Not a reserved keyword	`class_name`	`class`, `for`, `if`, `True`
Convention: meaningful snake_case for variables, CamelCase for classes, UPPER_CASE for constants	`avg_order_value`, `GST_RATE = 0.18`	`x1`, `a` (legal but unclear)

Data types

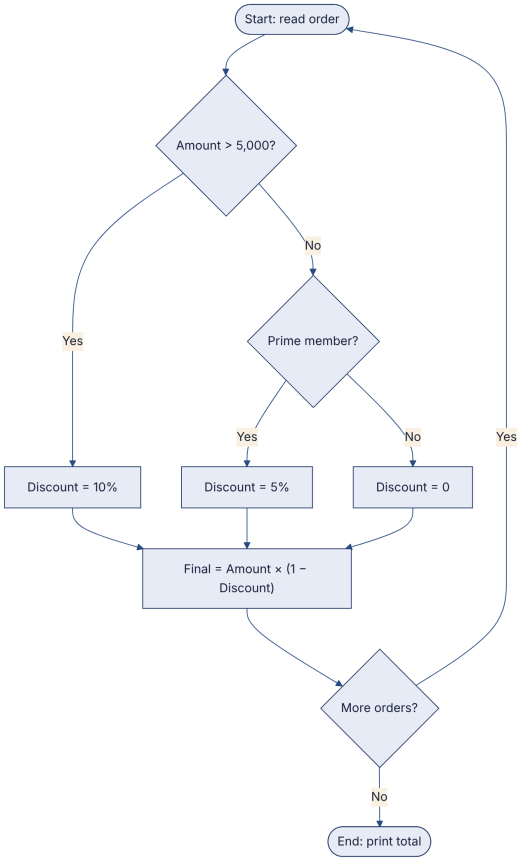
Integer	`42`	Counts, IDs
Float	`3.14`, `2499.0`	Prices, measurements
String	`"Gurugram"`	Text

Boolean	`True` / `False`	Flags, conditions
Collections	list `[1,2,3]`, tuple, dict `{"city":"Pune"}`, set	Groups of values
None / NULL	`None`	"No value"

Operators

Arithmetic	`+ - * / // % **` (// floor division, % remainder, ** power)
Comparison	`== != > < >= <=`
Logical	`and or not`
Assignment	`= += -= *=`
Membership / identity	`in`, `not in`, `is`

1.8 Control flow: conditions and loops



Flowchart of a discount rule with conditions and a loop over orders.

```
orders = [7200, 3100, 450]
prime = [False, True, False]
total = 0
for amount, is_prime in zip(orders, prime):      # loop: repeat for each order
    if amount > 5000:                             # condition
        discount = 0.10
    elif is_prime:
```

```
        discount = 0.05
    else:
        discount = 0
    total += amount * (1 - discount)
print(total)                                    # 6480 + 2945 + 450 = 9875.0

n = 0
while n < 3:                                    # while loop: repeat until condition is false
    n += 1
```

1.9 Functions and UDFs

A **function** is a named, reusable block that takes inputs (parameters) and returns an output. Built-in functions come with the language (`len` , `sum` , `round`); **UDFs** (user-defined functions) are ones you write.

```
def final_price(amount, is_prime=False):
    """Return the amount after the store's discount rule."""
    if amount > 5000:
        return amount * 0.90
    if is_prime:
        return amount * 0.95
    return amount

final_price(7200)          # 6480.0
final_price(3100, True)    # 2945.0
```

Benefits: write once, test once, reuse everywhere; easier to read and fix. Excel equivalent: `LAMBDA` ; SQL equivalent: user-defined functions and stored procedures.

1.10 Logic building and pseudo-code

Pseudo-code describes an algorithm in plain structured English before writing real code. It separates *thinking* from *syntax*.

```
PROBLEM: flag customers who have not ordered in 90 days

FOR each customer
    last_order = most recent order date for this customer
    IF no orders OR (today - last_order) > 90 days
        mark customer as "At risk"
    ELSE
        mark customer as "Active"
END FOR
OUTPUT count of At risk customers
```

1. Understand inputs and the exact output wanted.
2. Solve one example by hand.
3. Write the steps as pseudo-code / a flowchart.

4 Translate to code; test with normal and edge cases (empty list, zero, missing value).

Building Blocks ML · Class 2

Building Blocks ML · Class 2: Introduction to Statistics (Part 1/4)

Same lecture as **Building Blocks Essentials · Class 6**: what statistics is, population vs sample, data types and scales, and measures of central tendency. Revision summary:

- Descriptive statistics summarise; inferential statistics generalise from a sample to a population.
- Scales: nominal, ordinal, interval, ratio. Numeric data is discrete or continuous.
- Mean (symmetric data), median (skewed data), mode (categories), weighted mean (different quantities).

Practice

Q Classify each: PIN code, customer rating 1–5, temperature in °C, monthly spend.

PIN code: nominal (numbers used as labels). Rating: ordinal. Temperature °C: interval. Monthly spend: ratio.

Q Order values (₹): 450, 520, 480, 510, 9,800. Which average would you report and why?

Median = 510. The single ₹9,800 order pulls the mean to 2,352, which no typical customer spends.

Q A shop sells 100 units at ₹50 and 20 units at ₹200. What is the average selling price?

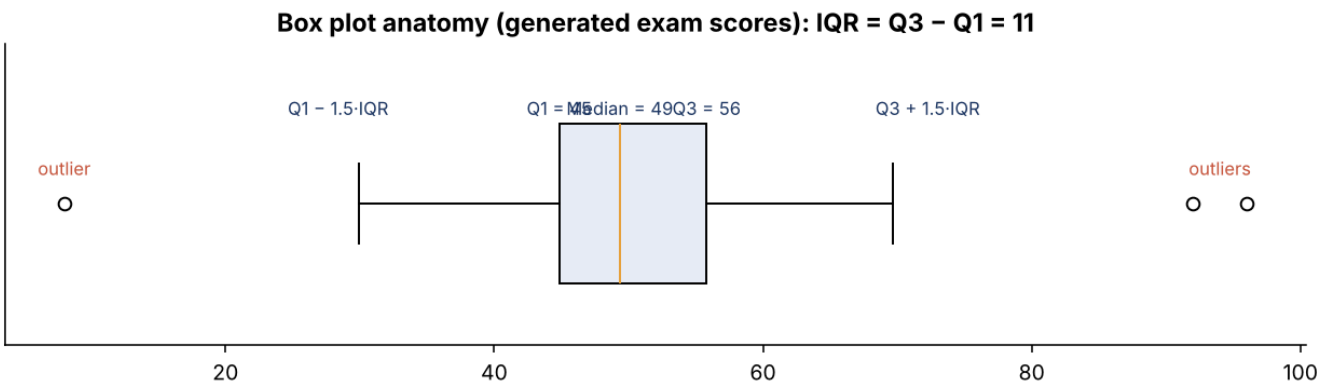
Weighted mean = (100×50 + 20×200) / 120 = 9,000 / 120 = ₹75, not the simple average of ₹125.

Python equivalents: `df["amt"].mean()`, `.median()`, `.mode()`, `np.average(x, weights=w)`.

Building Blocks ML · Class 3

Building Blocks ML · Class 3: Introduction to Statistics (Part 2/4)

Same lecture as **Building Blocks Essentials · Class 7**: range, variance, standard deviation, coefficient of variation, percentiles and quartiles, skewness and kurtosis, box plots and outliers, z-scores, covariance and correlation.



Box plot anatomy and the 1.5 × IQR outlier rule.

Practice

Q Data: 2, 4, 4, 4, 5, 5, 7, 9. Find the mean and population SD.

Mean = 40/8 = 5. Squared deviations: 9,1,1,1,0,0,4,16 = 32. Population variance = 32/8 = 4, SD = 2. (Sample SD = √(32/7) ≈ 2.14.)

Q Q1 = 40, Q3 = 60. Is 95 an outlier?

IQR = 20; upper fence = 60 + 1.5×20 = 90. 95 > 90, so yes.

Q r between ad spend and sales = 0.82. Can we say ads cause sales?

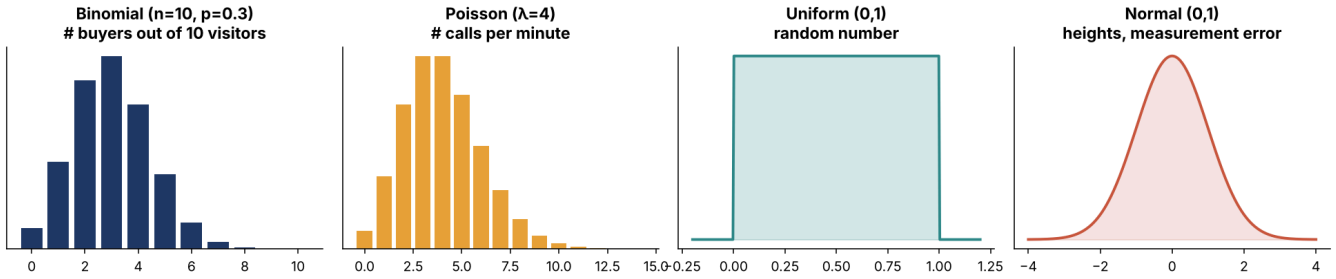
No. It is a strong positive association; causation needs an experiment (A/B test) or a careful causal analysis, because festive seasons may drive both.

Python: `df.describe()`, `df["x"].std()`, `df["x"].quantile([.25,.75])`, `df["x"].skew()`, `df.corr(numeric_only=True)`.

Building Blocks ML · Class 4

Building Blocks ML · Class 4: Introduction to Statistics (Part 3/4)

Same lecture as **Building Blocks Essentials · Class 8**: probability rules, conditional probability, Bayes, random variables, binomial/Poisson/normal distributions, expected value, sampling methods, sampling distribution and the CLT.



The distributions you will meet most often.

Practice

Q A call centre gets 4 calls per minute on average. Probability of exactly 6 calls in a minute?

Poisson: P(6) = $e^{-4} \cdot 4^6/6! \approx 0.104$. Excel `=POISSON.DIST(6,4,FALSE)`; Python `stats.poisson.pmf(6,4)`.

Q 30% of visitors buy. Probability that at least 1 of 5 visitors buys?

1 − P(none) = 1 − 0.7^5 = 1 − 0.168 = 0.832.

Q Heights ~ Normal(165, 8). What share is taller than 181 cm?

z = (181−165)/8 = 2 → about 2.3% (1 − 0.9772).

Q Population SD of order value is ₹600. What is the standard error of the mean for n = 100?

600/√100 = ₹60.

Building Blocks ML · Class 5

Building Blocks ML · Class 5: Introduction to Statistics (Part 4/4) – inferential statistics

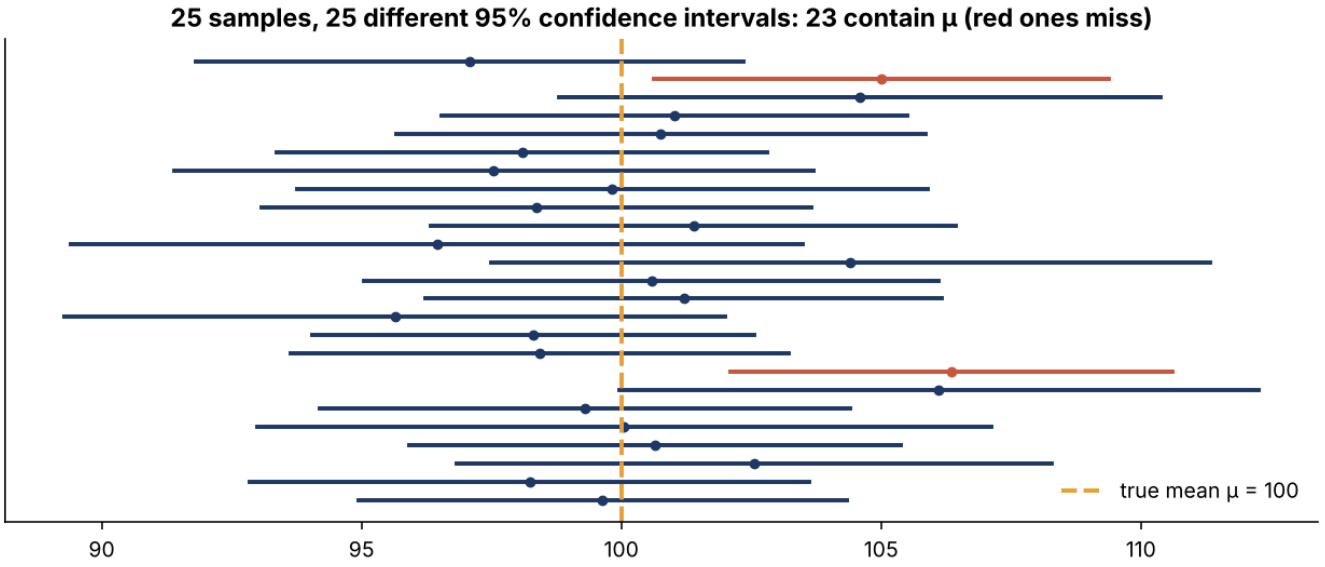
5.1 Point and interval estimation

A **point estimate** is a single best guess (sample mean $\bar{x}$ = ₹1,310). A **confidence interval** gives a range that captures the true value with a stated confidence:

```
CI = x̄ ± z* · s/√n   (z* = 1.96 for 95%; use t* for small samples)
```

Worked example

n = 400 orders, $\bar{x}$ = ₹1,310, s = ₹600. SE = 600/20 = 30. 95% CI = 1,310 ± 1.96×30 = **₹1,251 to ₹1,369**. Excel: `=CONFIDENCE.T(0.05,600,400)` returns the ± margin (≈ 59).

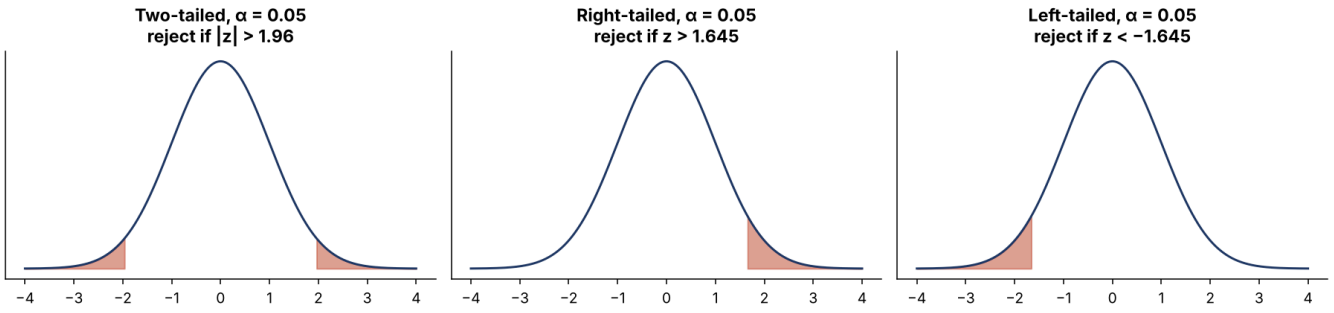


"95% confidence" describes the method: about 95 of every 100 intervals built this way contain the true mean (generated data).

Sample size for a margin E: $n = (z^* \cdot \sigma / E)^2$. To estimate average spend within ±₹50 with σ = 600: $n = (1.96 \times 600/50)^2 \approx 554$.

5.2 Hypothesis testing framework

1. H0: no change / no difference. H1: what you want evidence for.
2. Choose α (usually 0.05).
3. Pick the test; compute the statistic.
4. p-value ≤ α → reject H0; else fail to reject.
5. State the business conclusion and the effect size.



Two-tailed vs one-tailed rejection regions at α = 0.05.

Question	Test	Excel
Mean vs a target	One-sample z / t	`=T.TEST` needs two ranges; use `=T.DIST.2T(ABS(t),n-1)` on a computed t
Two independent groups	Two-sample t	`=T.TEST(A,B,2,3)` (two-tailed, unequal variance)
Before / after, same units	Paired t	`=T.TEST(Before,After,2,1)`
3+ groups	ANOVA	Data → Data Analysis → Anova: Single Factor
Two categorical variables	Chi-square	`=CHISQ.TEST(observed,expected)`
Two numeric variables	Correlation test / regression	Data Analysis → Regression

5.3 Errors

Type I (false positive, probability α): concluding a new checkout works when it does not. **Type II** (false negative, β): missing a real improvement. **Power** = 1 − β; increase it with a bigger sample.

5.4 Simple linear regression

$\hat{y} = b_0 + b_1 x$. Example: sales (₹ lakh) = 12 + 3.5 × ad spend (₹ lakh) means each extra lakh of ads is associated with ₹3.5 lakh more sales. $\mathbf{R^2}$ = share of variation in y explained by x. Excel: `=SLOPE()`, `=INTERCEPT()`, `=RSQ()`, `=FORECAST.LINEAR()`, or add a trendline to a scatter chart with "Display equation".

> **Tip** Full illustrated coverage of tests, p-values and errors: **ML Classes 1–2**; worked cases: **Hypothesis Testing case**.

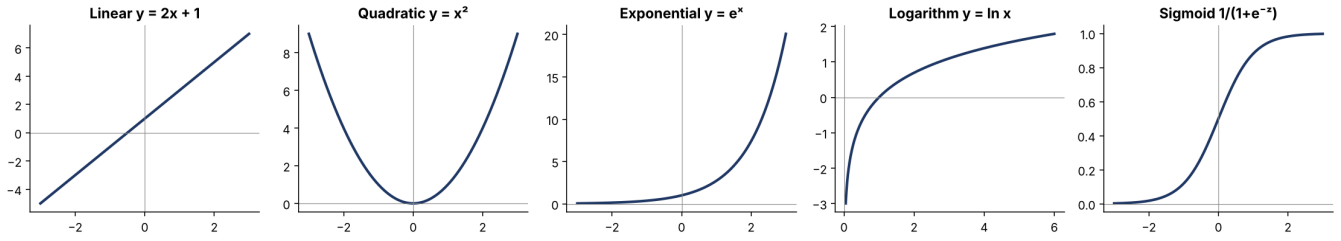
Building Blocks ML · Class 6

Building Blocks ML · Class 6: Introduction to Mathematics (Part 1/5) – algebra, functions, exponents and logs

6.1 Algebra refresher

- Variables and expressions: revenue = price × quantity → R = p·q.
- Solving linear equations: 3x + 5 = 20 → x = 5. Break-even: fixed cost ₹50,000, margin ₹250/unit → units = 50,000 / 250 = 200.
- Inequalities: 0.18x ≤ 9,000 → x ≤ 50,000.
- Summation notation: $\Sigma_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$ (Excel SUM). Product notation Π.

6.2 Functions and their shapes



Five shapes that appear throughout ML.

Linear y = mx + c	m = slope (change in y per unit x), c = intercept. Linear regression.
Quadratic y = ax^2 + bx + c	U-shape; squared error cost functions are quadratic, with one minimum.
Exponential y = a^{bx} / e^x	Compound growth; viral spread.
Logarithm y = log x	Inverse of exponential; compresses big ranges (income, population).
Sigmoid σ(z) = 1/(1+e^{-z})	Squashes any number to 0–1: logistic regression probabilities.

6.3 Exponents and logarithms

$a^m \cdot a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$, $a^{-n} = 1/a^n$	Exponent rules
log(ab) = log a + log b; log(a/b) = log a − log b; log(a^n) = n·log a	Log rules: turn products into sums
ln x = $\log_e$ x, e ≈ 2.718	Natural log used in ML and finance

Real-world use: CAGR

Revenue grew from ₹2 crore to ₹3.2 crore in 4 years. CAGR = (3.2/2)^(1/4) − 1 = 1.6^0.25 − 1 ≈ **12.5%**. Excel `=(B2/A2)^(1/4)-1` or `=RRI(4,2,3.2)`.

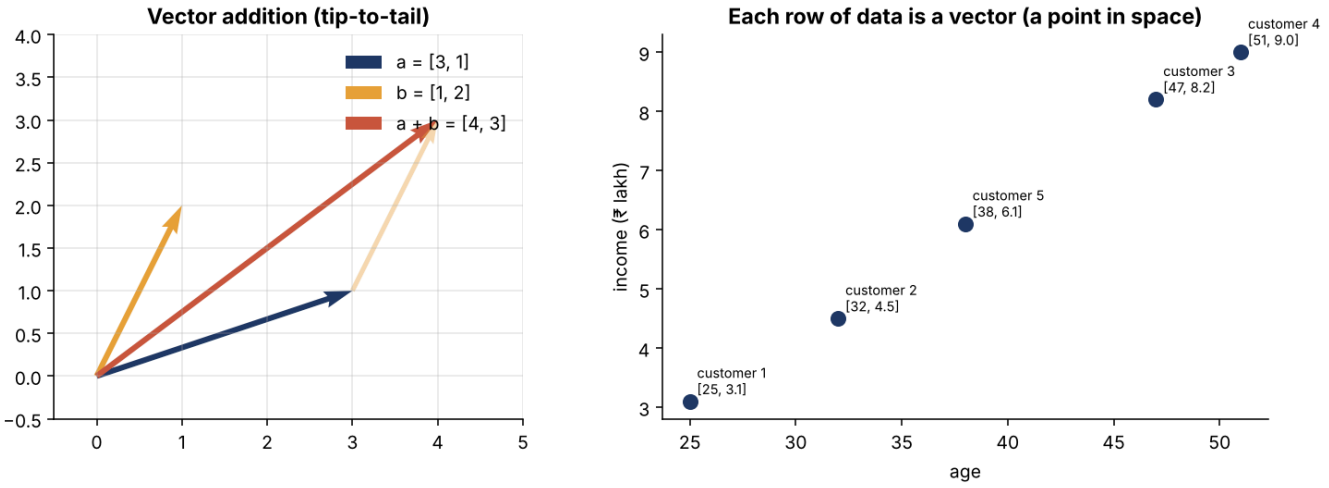
Why logs in data science: a log transform makes right-skewed data (income, sales) more symmetric; log-odds are the language of logistic regression; log-loss is a classification metric.

Building Blocks ML · Class 7

Building Blocks ML · Class 7: Introduction to Mathematics (Part 2/5) – vectors and matrices

7.1 Scalars, vectors, matrices, tensors

Scalar	A single number: 5
Vector	An ordered list: one customer = [age 32, income 4.5, visits 5]
Matrix	A 2-D grid: n customers × p features (a DataFrame without labels)
Tensor	3-D and higher: a batch of colour images = images × height × width × 3

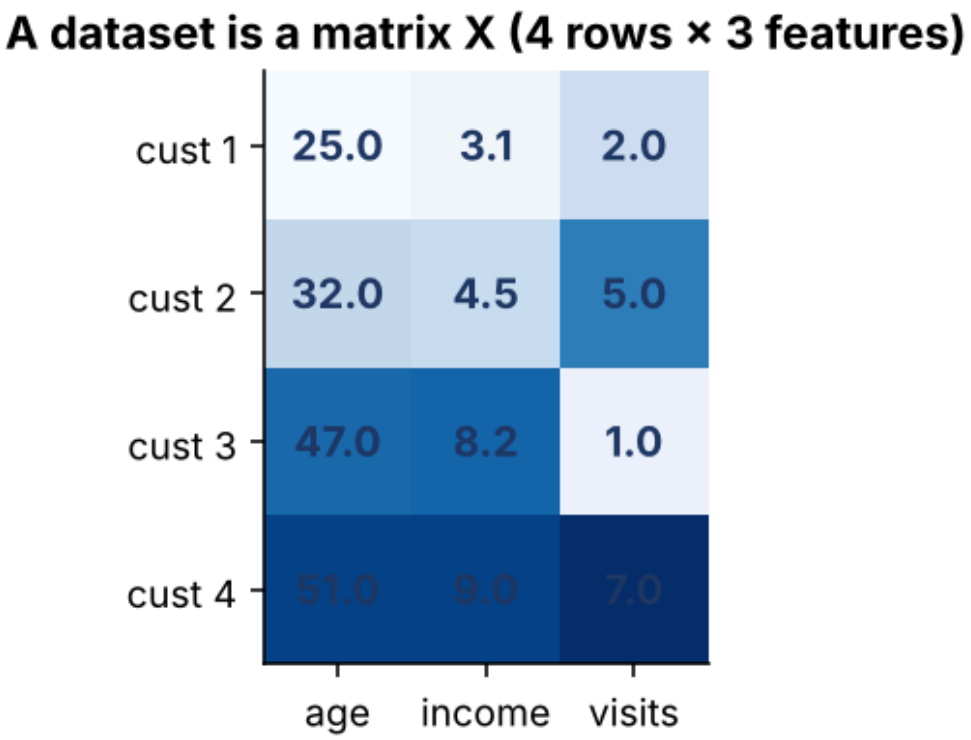


Left: adding vectors tip-to-tail. Right: each data row is a point (vector) in feature space; KNN and clustering measure distances between these points.

7.2 Vector operations

Operation	Example with a=[3,1], b=[1,2]	ML meaning
Addition	a + b = [4, 3]	Combining effects
Scalar multiplication	2a = [6, 2]	Scaling features
Dot product	a·b = 3×1 + 1×2 = 5	A linear model's prediction is a dot product: ŷ = w·x + b
Magnitude (norm)	‖a‖ = √(9+1) ≈ 3.16	Size of a vector; regularisation penalties
Euclidean distance	‖a − b‖ = √(4+1) ≈ 2.24	KNN, k-means
Cosine similarity	a·b / (‖a‖‖b‖) = 5/(3.16×2.24) ≈ 0.71	Text and embedding similarity (search, RAG)

7.3 Matrices



A dataset with 4 rows and 3 features is a 4 × 3 matrix X.

- Shape (rows × columns); element a_{ij} is row i, column j.
- Special matrices: square, identity I (1s on the diagonal), diagonal, symmetric ($A = A^T$), zero.
- Transpose A^T flips rows and columns.
- Addition and scalar multiplication work element by element (same shape required).

```
import numpy as np
X = np.array([[25, 3.1, 2], [32, 4.5, 5], [47, 8.2, 1], [51, 9.0, 7]])
X.shape        # (4, 3)
X.T.shape      # (3, 4)
a, b = np.array([3, 1]), np.array([1, 2])
a @ b          # 5  dot product
np.linalg.norm(a - b)   # 2.236 distance
```

Building Blocks ML · Class 8

Building Blocks ML · Class 8: Introduction to Mathematics (Part 3/5) – matrix operations

8.1 Matrix multiplication

(m × n) · (n × p) → (m × p). Inner sizes must match. Each output cell = dot product of a row of the first with a column of the second. Order matters: AB ≠ BA in general.

Real-world example: billing

Quantities Q (2 customers × 3 products) = [[2,1,0],[0,3,1]]; prices P (3 × 1) = [[500],[200],[1000]]. Bill = Q·P = [[2×500 + 1×200 + 0], [0 + 3×200 + 1×1000]] = **[[1,200],[1,600]]**. Excel: `=MMULT(Q,P)`. All predictions of a linear model at once: ŷ = Xw.

8.2 Determinant and inverse

- For A = [[a,b],[c,d]], det(A) = ad − bc. If det = 0 the matrix is **singular** (no inverse): its columns are redundant (perfect multicollinearity).
- Inverse A^{-1} satisfies $A \cdot A^{-1} = I$. For 2×2: $A^{-1} = (1/\det) \cdot [[d,-b],[-c,a]]$. Excel `=MINVERSE()`, `=MDETERM()`.

8.3 Systems of linear equations

2x + y = 1,100 and x + 3y = 1,800 (two baskets' totals) → A·[x,y] = b → [x,y] = $A^{-1}b$ = [300, 500]: item prices ₹300 and ₹500.

Linear regression's exact solution is the same idea: **$\mathbf{w = (X^T X)^{-1} X^T y}$** (the "normal equation").

```
A = np.array([[2, 1], [1, 3]]); b = np.array([1100, 1800])
np.linalg.solve(A, b)          # array([300., 500.])
np.linalg.det(A)               # 5.0
np.linalg.inv(A)
```

8.4 Eigenvalues and eigenvectors (intuition)

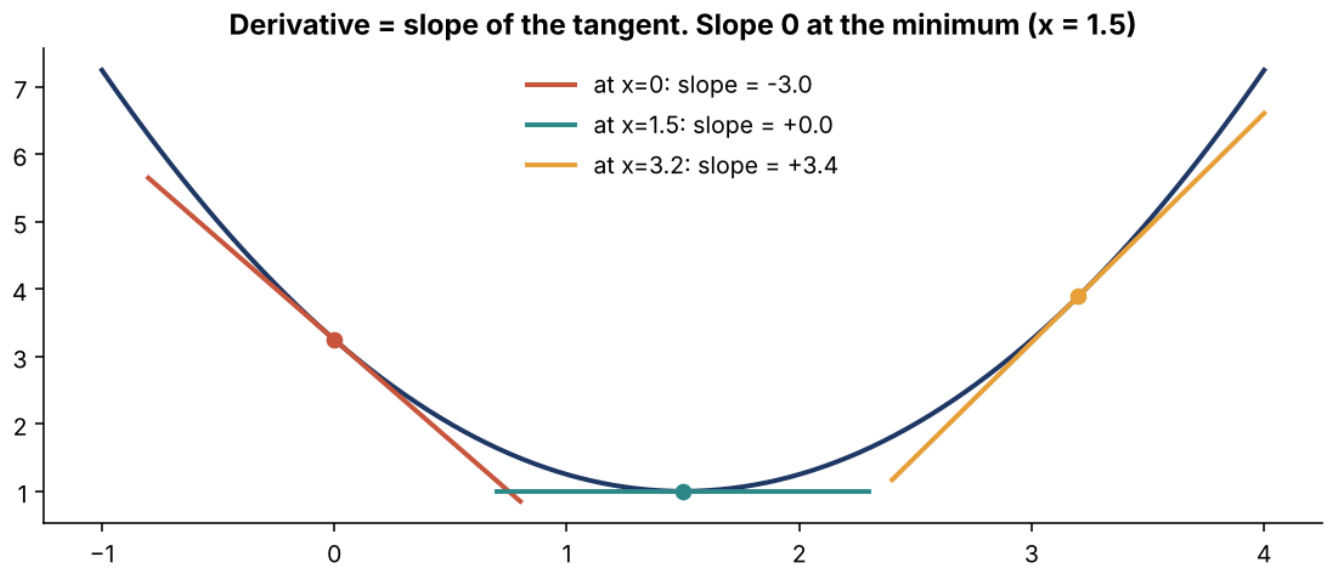
For a square matrix A, an eigenvector v keeps its direction when multiplied: Av = λv; λ is how much it is stretched. In data, the eigenvectors of the covariance matrix point along the directions of greatest spread, which is exactly what **PCA** uses to compress many correlated features into a few components.

Building Blocks ML · Class 9

Building Blocks ML · Class 9: Introduction to Mathematics (Part 4/5) – calculus for optimisation

9.1 Limits and the idea of a derivative

A derivative is the instantaneous rate of change: the slope of a curve at a point. f′(x) = lim(h→0) [f(x+h) − f(x)] / h.



Negative slope left of the minimum, zero at the minimum, positive to the right. Gradient descent always steps against the slope.

9.2 Differentiation rules

Rule	Formula	Example
Constant	$d/dx(c) = 0$	$d/dx(7) = 0$
Power	$d/dx(x^n) = n \cdot x^{n-1}$	$d/dx(x^3) = 3x^2$
Constant multiple / sum	$(c \cdot f)' = c \cdot f'$; $(f + g)' = f' + g'$	$d/dx(4x^2 + 3x) = 8x + 3$
Product	$(fg)' = f'g + fg'$	Revenue = $p \cdot q(p)$
Chain	$d/dx\ f(g(x)) = f'(g(x)) \cdot g'(x)$	$d/dx\ (2x+1)^2 = 2(2x+1) \cdot 2$
Exponential / log	$(e^x)' = e^x$; $(\ln x)' = 1/x$	Used in logistic regression
Sigmoid	$\sigma'(z) = \sigma(z)(1 - \sigma(z))$	Back-propagation in neural nets

9.3 Maxima, minima and optimisation

Business example: best price

Demand $q = 1{,}000 - 20p$. Revenue $R = p \cdot q = 1{,}000p - 20p^2$. $R' = 1{,}000 - 40p = 0 \rightarrow$ **p = ₹25**, q = 500, revenue ₹12,500. $R'' = -40 < 0$ confirms a maximum.

9.4 Partial derivatives and the gradient

With several inputs, a partial derivative $\partial f/\partial x$ changes one input while holding the others fixed. The **gradient** ∇f is the vector of all partials; it points uphill. For the regression cost $J(b_0,b_1) = (1/n)\Sigma(b_0 + b_1x - y)^2$:

$$\partial J/\partial b_0 = (2/n)\Sigma(\hat{y} - y) \qquad \partial J/\partial b_1 = (2/n)\Sigma(\hat{y} - y)\cdot x$$

Gradient descent update: $b \leftarrow b - \alpha\cdot\nabla J$. See it run on real numbers in **ML Class 5**.

9.5 Integrals (one idea)

An integral is the area under a curve: probabilities for continuous distributions are areas under the density curve ($P(Z \le 1.5) = 0.933$).

Building Blocks ML · Class 10

Building Blocks ML · Class 10: Introduction to Mathematics (Part 5/5) – probability and sets for ML

10.1 Sets

Union $A \cup B$ (either), intersection $A \cap B$ (both), complement A^c, difference $A - B$. Venn diagrams. Same ideas as SQL UNION / INTERSECT / EXCEPT and pandas merges.

10.2 Counting: permutations and combinations

Factorial n!	5! = 120 ways to order 5 items
Permutations nPr = n!/(n−r)!	Order matters: ways to pick 1st, 2nd, 3rd prize from 10 = 720
Combinations nCr = n!/(r!(n−r)!)	Order doesn't matter: ways to choose 3 features out of 10 = 120 (Excel `=COMBIN(10,3)`)

10.3 Conditional probability, independence, Bayes

$P(A|B) = P(A\cap B)/P(B)$. Independent if $P(A|B) = P(A)$. Bayes: $P(A|B) = P(B|A)P(A)/P(B)$, the basis of **Naive Bayes** spam filters: P(spam | "free offer") from how often "free offer" appears in spam vs genuine mail.

10.4 Random variables, expectation, variance

- $E[X] = \Sigma\, x\cdot P(x)$ (discrete) — expected profit of a campaign.
- $Var(X) = E[X^2] - (E[X])^2$.
- Joint and marginal distributions; covariance between two random variables.

Real-world example: expected value of a decision

A retention offer costs ₹200. If offered, 30% of at-risk customers stay, each worth ₹2,000 → expected gain = 0.3×2,000 − 200 = **₹400 per customer**. Positive, so offer it. Churn models supply the probabilities.

10.5 Likelihood and maximum likelihood (intuition)

Likelihood asks: given these parameters, how probable is the data we actually saw? Logistic regression picks coefficients that make the observed 0/1 outcomes most likely; minimising **log-loss** is the same thing.

Interview Q&A

Q Compiler vs interpreter?

A compiler translates the whole program into machine code before running (fast execution, all errors upfront); an interpreter translates and runs line by line (instant feedback, stops at the first error). Python is interpreted (via bytecode).

Q What is pseudo-code and why use it?

Plain-language step-by-step logic written before coding. It lets you design and check the algorithm without worrying about syntax.

Q What is a confidence interval?

A range built from sample data that, by the method used, captures the true population value a stated share of the time (e.g. 95%).

Q Why is the dot product important in ML?

A linear model's prediction, a neural-network neuron's input, and cosine similarity between embeddings are all dot products.

Q What does a zero determinant mean for regression?

X^TX is not invertible because some features are exact linear combinations of others (perfect multicollinearity), so coefficients cannot be uniquely estimated.

Q Why do we need derivatives in ML?

To find which direction to change parameters to reduce error; gradient descent uses the gradient of the cost function.

Q Explain Bayes' theorem with an example.

It updates a prior belief with evidence. With 1% fraud and a test that is 95% sensitive with a 5% false-alarm rate, only about 16% of flagged transactions are fraud.

Resources

- ▶ YouTube: Simplilearn programming basics for beginners · ▶ YouTube: CS50 lecture 0 scratch computational thinking

- ▶ YouTube: Simplilearn statistics for data science · ▶ YouTube: StatQuest confidence intervals clearly explained
- ▶ YouTube: 3Blue1Brown essence of linear algebra · ▶ YouTube: 3Blue1Brown essence of calculus
- ▶ YouTube: Khan Academy linear algebra · ▶ YouTube: Khan Academy probability and combinatorics
- ▶ YouTube: Simplilearn mathematics for machine learning
- 📘 Mathematics for Machine Learning (free book, Deisenroth et al.)
- 📘 Python official tutorial